



**K.L.E. SOCIETY'S
CHIDANAND B. KORE POLYTECHNIC,
CHIKODI-591201**

District : Belagavi



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2024-2025

A Internship Report On

“OJT 1 -- ADVOCAI”

“OJT 2 -- ONLINE GROCERY STORE”

In partial fulfilment for the award of the diploma of

DIPLOMA IN COMPUTER SCIENCE AND ENGINEERING

SUBMITTED BY

Mr. AJIT VITTHAL ALAGUNDI

DTE REG : 339CS22006

Under the guidance of

Mr. Anoop Desai

**Designation - Full Stack Developer,
Pranaksh Technologies LLP,
Belagavi**



**K.L.E. SOCIETY'S
CHIDANAND B. KORE POLYTECHNIC,
CHIKODI-591201**

District : Belagavi



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2024-2025

A Internship Report On

“OJT 1 -- ADVOCAT”

“OJT 2 -- ONLINE GROCERY STORE”

In partial fulfilment for the award of the diploma of

DIPLOMA IN COMPUTER SCIENCE AND ENGINEERING

SUBMITTED BY

Mr. AJIT VITTHAL ALAGUNDI

DTE REG : 339CS22006

Under the guidance of

Mr. Anoop Desai

Designation - Full Stack Developer,
Pranaksh Technologies LLP,
Belagavi

Internship Guide
Mr. Anoop Desai

Cohort Owner
Prof. S.S Khot

Head of Department
Prof. S.K Tavadare

Principal
Prof. D.D Billur



**K.L.E. SOCIETY'S
CHIDANAND B. KORE POLYTECHNIC,
CHIKODI-591201**

District : Belagavi



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2024-2025

CERTIFICATE

This is to certify that the Internship report entitled “**ADVOCAI & ONLINE GROCERY STORE**” was carried out by **Mr. AJIT VITTHAL ALAGUNDI** of **6th** Semester Diploma in Computer Science and Engineering, **K.L.E. Society's C. B. Kore Polytechnic, Chikodi**. In practical fulfillment of the Diploma in **Computer Science and Engineering** award requirement during **2024 – 2025**. This work has been carried out under the guidance of **Mr. Anoop Desai**, **Designation - Full Stack Developer, Pranaksh Technologies LLP, Belagavi**. The Internship work prescribed for the said Diploma.

It is further understood that by this certificate the undersigned does not endorse or approve any statement made, opinion expressed, or conclusion purpose for which it is submitted.

Internship Guide
Mr. Anoop Desai

Cohort Owner
Prof. S.S Khot

Head of Department
Prof. S.K Tavadare

Principal
Prof. D.D Billur



**K.L.E. SOCIETY'S
CHIDANAND B. KORE POLYTECHNIC,
CHIKODI-591201**

District : Belagavi



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2024-2025

STUDENT DECLARATION

We, **Mr. AJIT VITTHAL ALAGUNDI** A Sixth-Semester Diploma in Computer Science and Engineering students of **K.L.E Society's Chidanand B. Kore Polytechnic, Chikodi**. Hereby declare that the work entitled "**Advoc AI & Online Grocery Store**" has been carried out independently by us under the guidance **Mr. Anoop Desai, Designation - Full Stack Developer, Pranaksh Technologies LLP, Belagavi**. we further declare that the work reported in this Internship has not been submitted and will be submitted in part or full for the award of any after diploma or degree in this institute or any other institute or university.

Mr. Ajit Vitthal Alagundi (339CS22006)

Date :

Place : Chikodi

INTERNSHIP CERTIFICATE ISSUED BY THE ORGANIZATION

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to all those who have supported and guided me throughout the successful completion of this Internship.

First and foremost, I would like to thank Prof. DARSHANKUMAR D. BILLUR, Principal of K.L.E. SOCIETY'S C.B. KORE POLYTECHNIC CHIKODI-591201, for providing the necessary facilities and a motivating environment to carry out this work.

I express my deep sense of gratitude to Prof. SANTOSH K TAVADARE, Head of the Department of COMPUTER SCIENCE AND ENGINEERING, for their constant encouragement and valuable support during the course of this Internship.

I am especially thankful to my Internship guide, Mr. ANOOP DESAI, Designation - Full stack Developer for their consistent guidance, suggestions, and valuable feedback that helped shape this Internship into its final form.

I would also like to thank all the faculty members and staff of the PRANAKSH TECHNOLOGIES LLP, BELAGAVI. for their help and cooperation throughout the duration of this Internship.

I extend my heartfelt thanks to all the Staff Members of the Department of COMPUTER SCIENCE AND ENGINEERING for their continuous support and assistance throughout my academic journey.

Last but not the least, I am deeply thankful to my family and friends for their unwavering support, patience, and encouragement during every stage of this journey.

EXECUTIVE SUMMARY

On Job Training1: AdvocaAI

AdvocaAI is an AI-powered advocacy platform developed to assist users with information and support on social issues. It uses intelligent chatbots to automate interactions and improve accessibility.

The motivation behind this project is to combine technology with social good by making advocacy more accessible and responsive through AI.

The development process involved literature review, requirement analysis, tool selection, prototyping, and user testing.

Key results include a functional chatbot, increased user engagement, and improved platform accessibility. The system shows strong potential for real-world use in advocacy and public support.

On Job Training 2: Online Grocery Store

The Online Grocery Store project focuses on building an AI-enhanced e-commerce platform to improve the online shopping experience.

It was inspired by the rising need for convenient grocery shopping solutions and the growing role of AI in e-commerce.

The team worked on system design, frontend and backend development, and testing based on user feedback.

Major outcomes include AI-powered product recommendations, faster load times, and improved customer satisfaction. The platform offers a practical and efficient solution for modern grocery shopping.

CONTENTS

[OJT – 1 ADVOC AI]

SL No	Contents	Page No
1	Abstract	1-1
2	Introduction	2-2
3	Existing System and Literature Survey 1. Existing System 2. Literature Survey	3-7
4	Problem Statement and Objectives 1. Problem Definition 2. Challenges 3. Limitations 4. Applications	8-8
5	Requirements 1. Hardware Requirements 2. Software Requirements 3. Functional Requirements 4. Non-Functional Requirements	9-10
6	System Design and Methodology 1. Methodology 2. Software Architecture 3. Database Design 4. UI Design 5. Module Design 6. ERD Diagram	11-14
7	Pseudocode	15-18
8	Experiments and Results	19-21
9	Test Cases	22-26
10	Future Scope	27-28
11	Conclusion	29-29
12	References	30-30

CONTENTS

[OJT -2 ONLINE GROCERY STORE]

SL No	Contents	Page No
1	Abstract	1-1
2	Introduction	2-2
3	Existing System and Literature Survey 1. Existing System 2. Literature Survey	3-3
4	Problem Statement and Objectives 1. Problem Definition 2. Challenges 3. Limitations 4. Applications	4-4
5	Requirements 1. Hardware Requirements 2. Software Requirements 3. Functional Requirements 4. Non-Functional Requirements	5-5
6	System Design and Methodology 1. Methodology 2. Software Architecture 3. Database Design 4. UI Design 5. Module Design 6. ERD Diagram	6-7
7	Experiments and Results	8-15
8	Test Cases	16-20
9	Project Core Features	21-22
10	Future Scope	23-23
11	Conclusion	24-24
12	References	25-25

LIST OF FIGURES

Figure No.	Title of the Figure	Page No.
2.1	System Architecture Diagram	11
2.2	ERD Diagram	14
2.3	Screenshot of Environment setup	19
2.4	Screenshot of Register new user	20
2.5	Screenshot of Login Dashboard	20
2.6	Screenshot of Homepage	21
2.7	Screenshot of Prompt	21
3.1	ERD Diagram (Level 1)	8
3.2	Screenshot of Environment setup	9
3.3	Screenshot of Homepage	9
3.4	Screenshot of Register new user	10
3.5	Screenshot of Login Dashboard	10
3.6	Screenshot of Home Pages	11,12
3.7	Screenshot of Selecting Product	12
3.8	Screenshot of Adding To Cart	13
3.9	Screenshot of Payment Of The Product	13
3.10	Screenshot of Order Has Been Successful	14
3.11	Screenshot of Order History	14
3.12-15	Screenshots of Setting	15-16

LIST OF TABLES

Table No.	Title of the Table	Page No.
2.1	Literature Survey	7
2.2	Software and Hardware Requirements	9
3.1	Test Cases and Expected Outcomes	17
3.2	Comparison Between Existing and Proposed System	22
3.3	Performance Evaluation Results	23

ABBREVIATIONS / NOTATIONS / NOMENCLATURE

Abbreviation / Symbol	Full Form / Description
IoT	Internet of Things
API	Application Programming Interface
GUI	Graphical User Interface
CPU	Central Processing Unit
RAM	Random Access Memory
HTML	Hyper Text Markup Language
DFD	Data Flow Diagram
UML	Unified Modeling Language
DBMS	Database Management System
SQL	Structured Query Language
HTTP	Hyper Text Transfer Protocol
IP	Internet Protocol
JSON	JavaScript Object Notation
Mbps	Megabits per second

CHAPTER-1

COMPANY PROFILE

Pranaksh Technologies, established in 2023, is a dynamic company specializing in technology solutions, digital marketing, and academic training. Initially focused on educational technology services for IT companies, the firm has expanded its expertise to include software development, artificial intelligence, embedded systems, cybersecurity, and the Internet of Things (IoT).

Vision and Mission

Pranaksh Technologies aims to be a trusted leader in delivering high-quality services that bridge the gap between academia and the corporate world. Their mission is to provide exceptional solutions and training that connect academic learning with corporate success, empowering individuals and organizations with cutting-edge skills and knowledge.

Services Offered

- **Corporate Solutions:** Offering smart strategies and tools to boost efficiency, tackle challenges, and drive growth, ensuring success in today's competitive market.
- **Academic Training:** Providing hands-on expertise in fields like AI and Data Science, equipping individuals with the skills needed to stay competitive in a rapidly evolving landscape.

Technology Focus

- Internet of Things (IoT)
- Data Science
- Artificial Intelligence & Machine Learning
- Web Development

Milestones

Since its inception, Pranaksh Technologies has achieved significant milestones:

- **50+** Happy Clients
- **100+** Students Trained
- **150+** Projects Completed
- **10,000+** Training Hours Delivered

These accomplishments reflect the company's dedication to excellence and client satisfaction.

Why Choose Pranaksh Technologies

The company's value-driven approach ensures cost-effective services without compromising quality. They emphasize innovative partnerships, working closely with clients to deliver creative solutions. With a focus on trust and tangible results, Pranaksh Technologies prepares individuals with corporate-ready skills through programs that mirror real-world expectations.

Contact Information

Pranaksh Technologies is located in Belagavi, Karnataka. For inquiries:

- **Email:** info@pranakshtechnologies.com
- **Phone:** +91-8867635488

For more detailed information, please visit their official website: <https://pranakshtech nologies.com>

CHAPTER-2

ON JOB TRAINING – 1 “ADVOCAI”

ABSTRACT

AdvocAI is an AI-powered legal advisory platform developed to offer cost-effective and accessible legal guidance, specifically tailored to the Indian legal system. Legal consultations in traditional settings often involve high fees, long wait times, and limited access—particularly for those in rural or underserved areas. AdvocAI seeks to bridge this gap by providing digital legal assistance that is both affordable and easy to use.

The platform utilizes advanced natural language processing (NLP) and machine learning algorithms to analyze legal documents and accurately respond to user queries. These technologies help decode complex legal language and deliver relevant, understandable information to users, enabling them to make informed decisions without needing a legal background. This AI-driven approach ensures faster, more consistent support while reducing dependency on in-person consultations.

AdvocAI offers a user-friendly interface that includes secure authentication and conversational AI tools, allowing users to interact naturally with the system while maintaining data privacy. Whether someone is seeking advice on contracts, property laws, or civil matters, the platform provides tailored responses in real-time. It is designed to accommodate both general queries and document-based input, making it versatile and practical for a wide range of legal needs.

Continuously evolving through user feedback and legal updates, AdvocAI aims to democratize legal knowledge by reducing barriers to expert consultation. The platform emphasizes accuracy, security, and compliance with the latest legal standards, ensuring reliable support for its users. Ultimately, AdvocAI represents a transformative step toward making legal advice more inclusive, empowering individuals with the tools they need to understand and act on there.

INTRODUCTION

AdvocAI is an innovative AI-driven legal advisor designed to deliver precise legal advice grounded in Indian legal frameworks. The primary aim of this project is to bridge the accessibility gap in legal knowledge, making it available to a broader audience without the necessity for costly legal consultations. By utilizing advanced AI technologies, AdvocAI processes legal documents and responds to user queries, ensuring individuals can navigate complex legal issues effectively. The platform's user-friendly design further enhances its accessibility, enabling users to obtain accurate and timely legal advice, potentially revolutionizing how legal services are accessed in India.

The significance of AdvocAI lies in its ability to democratize legal knowledge. Traditional legal consultations can be expensive, and many individuals may not have the financial resources to access the legal advice they need. AdvocAI addresses this issue by providing a cost-effective solution that reduces reliance on expensive legal services. By integrating natural language processing (NLP) and machine learning algorithms, the platform can understand and interpret legal jargon, offering users reliable advice tailored to their specific inquiries. Over time, the system improves based on user interactions and feedback, ensuring that the quality of advice remains high.

In addition to its affordability, AdvocAI is designed with user-friendliness in mind. The platform is intuitive and easy to navigate, making it accessible to individuals with varying levels of legal knowledge. This user-centric approach ensures that even those unfamiliar with legal terminology can confidently use the platform to address their legal concerns.

Overall, AdvocAI represents a significant step forward in making legal advice more accessible and affordable. By leveraging advanced AI technologies, the platform empowers individuals to navigate the complexities of the legal system, fostering a more informed and empowered society. This project underscores the transformative potential of AI in the legal domain, highlighting the importance of technological innovation in enhancing public access to legal resources.

EXISTING SYSTEM AND LITERATURE SURVEY

Existing System

Prior to the launch of AdvocAI, individuals in need of legal advice typically relied on traditional avenues, such as meeting with lawyers in person. These consultations could be costly and time-consuming, making it difficult for many to access quality legal support. Although legal aid services are available for low-income individuals, they are often limited in scope and can be challenging to navigate. Additionally, online legal services might offer basic advice but frequently lack the personalization needed to address individual concerns, often leading to confusion due to the use of complex legal jargon.

Many people also sought guidance from online forums or community groups, where they could pose questions and receive feedback from others. However, the reliability of the advice found in these spaces is often uncertain, which can result in misinformation. Others turned to legal textbooks and online materials, but such resources are often dense and difficult for those without a legal background to understand, further hindering access to crucial legal information.

These challenges highlight the urgent need for a more effective and accessible solution. AdvocAI addresses this gap by harnessing advanced AI technologies to offer reliable and affordable legal guidance. With its user-friendly approach, AdvocAI empowers individuals to confidently navigate the complexities of the legal system, making legal knowledge more accessible to all.

Literature Survey

1. Legal Judgment Prediction: Datasets, Metrics, Models, and Challenges Luyang Han, Kunlin Yang, Zhaoyu Zhang, (2022)

In the paper "A Survey on Legal Judgment Prediction: Datasets, Metrics, Models, and Challenges," the authors review various approaches to legal judgment prediction (LJP) using machine learning. They highlight the importance of LJP in assisting legal professionals by predicting court decisions based on case facts. Key aspects discussed include:

- **Datasets:** The availability of high-quality datasets is crucial for developing effective LJP models. The paper reviews several publicly available datasets from different jurisdictions, emphasizing the need for more comprehensive and diverse datasets to improve model generalizability.
- **Metrics:** Various metrics are used to evaluate LJP models, including accuracy, F1 score, and area under the curve (AUC). The paper suggests that while accuracy is a common metric, it is not always the best indicator of a model's performance in real-world applications.
- **Models:** The survey covers a range of machine learning models used in LJP, from traditional approaches like logistic regression to more advanced techniques such as deep learning. The authors note the growing trend of using neural networks due to their ability to capture complex patterns in legal texts.
- **Challenges:** The paper outlines several challenges in the field, including the complexity of legal language, the need for interpretability of models, and ethical concerns related to fairness and bias.

2. Legal Guidance through Artificial Intelligence: Chatbot for Sexual Violence Victims and Survivors

Sung Yoon Whang, Byung Ha Kim, Sung Ju Hwang, Bomi Kim

(2021)

The paper "Legal Guidance Through Artificial Intelligence: Chatbot for Sexual Violence Victims and Survivors" discusses the development of an AI chatbot designed to provide legal assistance to sexual violence victims. Key points include:

- **Need for Legal Guidance:** The paper highlights the difficulties victims face in accessing legal help, particularly due to stigma, fear, and lack of awareness. An AI chatbot offers a discreet and accessible solution.
- **Chatbot Development:** The chatbot utilizes natural language processing (NLP) to understand user queries and provide relevant legal information. It is designed to be user-friendly and empathetic, ensuring that users feel supported throughout the interaction.
- **Implementation:** The authors discuss the implementation details, including the choice of language models, integration with legal databases, and continuous improvement through user feedback.
- **Impact and Limitations:** While the chatbot provides valuable support, the paper acknowledges limitations such as the need for regular updates to legal content and the potential for misuse or misunderstanding of the information provided.

3. Preserving Privacy in Arabic Judgments: AI-Powered Anonymization for Enhanced Legal Data Privacy

Asim Karim, Muhammad Arif Butt, Irum Inam, Sajid Ali

(2022)

In "Preserving Privacy in Arabic Judgments: AI-Powered Anonymization for Enhanced Legal Data Privacy," the authors explore methods to protect the privacy of individuals in legal documents through AI-driven anonymization. Key aspects include:

- **Privacy Concerns:** The paper addresses the importance of anonymizing legal documents to protect sensitive information and comply with privacy laws, particularly in the context of Arabic legal texts.
- **Anonymization Techniques:** The study reviews various anonymization techniques, including rule-based methods and machine learning approaches. It highlights the effectiveness of AI in identifying and masking personal information while preserving the document's utility for legal purposes.
- **Evaluation:** The authors propose evaluation metrics to assess the performance of anonymization techniques, focusing on the balance between privacy protection and data utility.
- **Challenges and Future Work:** The paper identifies challenges such as the complexity of legal language and the need for context-aware anonymization. Future research directions include improving the accuracy of anonymization models and addressing ethical considerations related to data privacy.

Title	Focus Area	Datasets	Metrics	Models	Challenges
A Survey on Legal Judgment Prediction: Datasets, Metrics, Models, and Challenges	Legal Judgment Prediction (LJP)	Public datasets from various jurisdictions	Accuracy, F1 score, AUC	Logistic regression, deep learning	Legal language complexity, model interpretability, bias
Legal Guidance Through AI: Chatbot for Sexual Violence Victims	AI chatbot for legal help	Not specified	Not specified	NLP models	Regular updates, misuse risk
Preserving Privacy in Arabic Judgments: AI-Powered Anonymization for Enhanced Legal Data Privacy	AI-driven anonymization to protect privacy in legal documents	Not specified	Balance between privacy protection and data utility	Rule-based methods and machine learning approaches	Complexity of legal language, context-aware anonymization, ethical considerations

Table 2.1 Literature Survey

PROBLEM STATEMENT AND OBJECTIVES

Problem Definition/Aim of the Project

Access to legal consultation is often constrained by high costs and a shortage of legal professionals, particularly in remote areas. This creates a pressing need for a cost-effective, reliable, and easily accessible solution that can provide legal advice rooted in Indian laws. The project Advoc AI aims to address this gap by leveraging artificial intelligence to deliver expert legal guidance through an intuitive digital platform.

Challenges

- **High Costs:** Traditional legal consultations are often financially prohibitive for many individuals.
- **Geographical Barriers:** Limited availability of legal professionals in remote or rural areas restricts access to legal advice.
- **Complex Legal Terminology:** Users may struggle to understand legal jargon, making it difficult to navigate their legal concerns.
- **Ensuring Accuracy:** Maintaining the reliability and accuracy of legal advice generated by AI systems presents a significant challenge.

Limitations

- **AI Limitations:** While AI can provide valuable insights, it may not fully replicate the nuanced understanding of a human legal expert.
- **Dependence on Technology:** Users with limited access to technology or the internet may struggle to benefit from the platform.
- **Legal Jurisdiction:** The platform must ensure that the legal advice provided is applicable and relevant to specific Indian laws and regulations.

Applications

- **Personal Legal Advice:** Individuals seeking guidance on personal legal matters, such as contracts, property disputes, and family law issues.
- **Educational Resource:** Serving as a tool for users to learn about their legal rights and responsibilities in a simplified manner.

REQUIREMENTS

Hardware Requirements

- **Processor:** Intel Core i5 or equivalent
- **RAM:** 8 GB
- **Storage:** 100 GB HDD or SSD
- **Internet:** Reliable internet connection for API access and updates

Software Requirements

- **Operating System:** Windows 10 or higher, macOS, or Linux
- **Programming Language:** Python 3.8 or higher
- **Libraries and Packages:**
 - PyPDF2 for PDF processing
 - Lang chain for language model interactions
 - Lang chain google-generative ai for Google Generative AI embedding's
 - google-generative ai for API interaction
 - lang chain community for community-supported extensions
 - python-dotenv for environment variable management
 - streamlit for web interface
 - faiss-cpu for vector storage and search
 - Streamlit authenticator for user authentication
- **Database:** SQLite for user and chat history management

Functional Requirements

- **User Authentication:**
 - User registration with profile image upload.
 - Secure login and logout functionality.
- **Document Processing:**
 - Ingest and process PDF documents to extract text.
- **Text Chunking:**
 - Split extracted text into chunks suitable for embedding and analysis.

➤ **Vector Store Creation:**

- Embed text chunks and store them in a FAISS vector store for efficient retrieval.

➤ **Conversational AI:**

- AI-powered chat interface for legal queries.
- Responses generated using context from processed documents.

➤ **Session Management:**

- Start new chat sessions and access previous conversations.
- Clear chat history functionality

Non-Functional Requirements

➤ **Performance:**

- Fast response times to user queries.

➤ **Scalability:**

- Ability to handle multiple concurrent users.

➤ **Security:**

- Ensure the privacy and security of user data and interactions.

➤ **Usability:**

- Intuitive and accessible user interface

SYSTEM DESIGN AND METHODOLOGY

Methodology

The development of AdvocAI will follow an Agile methodology, allowing for iterative progress through sprints. This approach facilitates frequent feedback from stakeholders, enabling the team to make necessary adjustments and improvements. Key phases will include requirement gathering, design, development, testing, and deployment, ensuring that the platform meets user needs effectively.

Software Architecture

The software architecture of AdvocAI will be based on a client-server model, where the client (user interface) interacts with the server (backend logic and database). The architecture will consist of the following layers:

Presentation Layer: Built using Streamlit, this layer will provide a user-friendly interface for interactions.

Application Layer: This layer will handle the business logic, including user authentication, document processing, and query handling through AI.

Data Layer: The database (SQLite) will manage user data, chat history, and processed documents, ensuring efficient retrieval and storage.

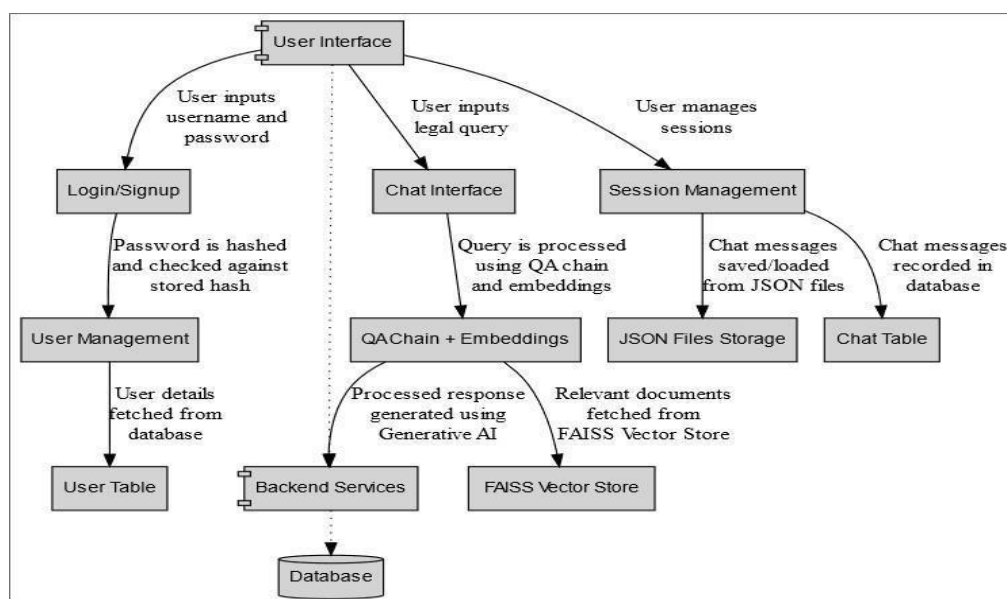


Figure 2.1 Software Architecture

Database Design

The database design for Advoc AI will include the following key tables: Users Table:

- Fields: User ID (Primary Key), Username, Password, Profile Image, Registration Date. Chat History Table:
- Fields: Chat ID (Primary Key), User ID (Foreign Key), Chat Time stamp, Chat Content Documents Table:
- Fields: Document ID (Primary Key), User ID (Foreign Key), Document Name, Processed Text, Upload Date.

UI Design

- The user interface will focus on simplicity and accessibility. Key features will include:
- Login/Register Page: A straightforward interface for user authentication, including image upload functionality.
- Dashboard: A central hub where users can initiate new chat sessions, view chat history, and upload documents.
- Chat Interface: An interactive space for users to engage with the AI, displaying responses alongside input fields.
- Help/FAQ Section: A dedicated area for users to access educational resources and FAQs related to legal advice and platform use.

Module Design

AdvocAI will be divided into several key modules, each responsible for specific functionalities:

User Management Module:

- Handles user registration, login, and authentication.
- Manages user profiles, including profile image uploads.

Document Processing Module:

- Ingests and processes PDF documents.
- Extracts text and prepares it for analysis.

Conversational AI Module:

- Facilitates interaction between users and the AI system.
- Generates responses based on user queries and processed documents.

Session Management Module:

- Manages user chat sessions, allowing access to previous conversations.
- Provides functionality to clear chat history.

Feedback Module:

- Collects user feedback on the AI's performance.
- Implements continuous improvement based on user input.

ERD Diagram

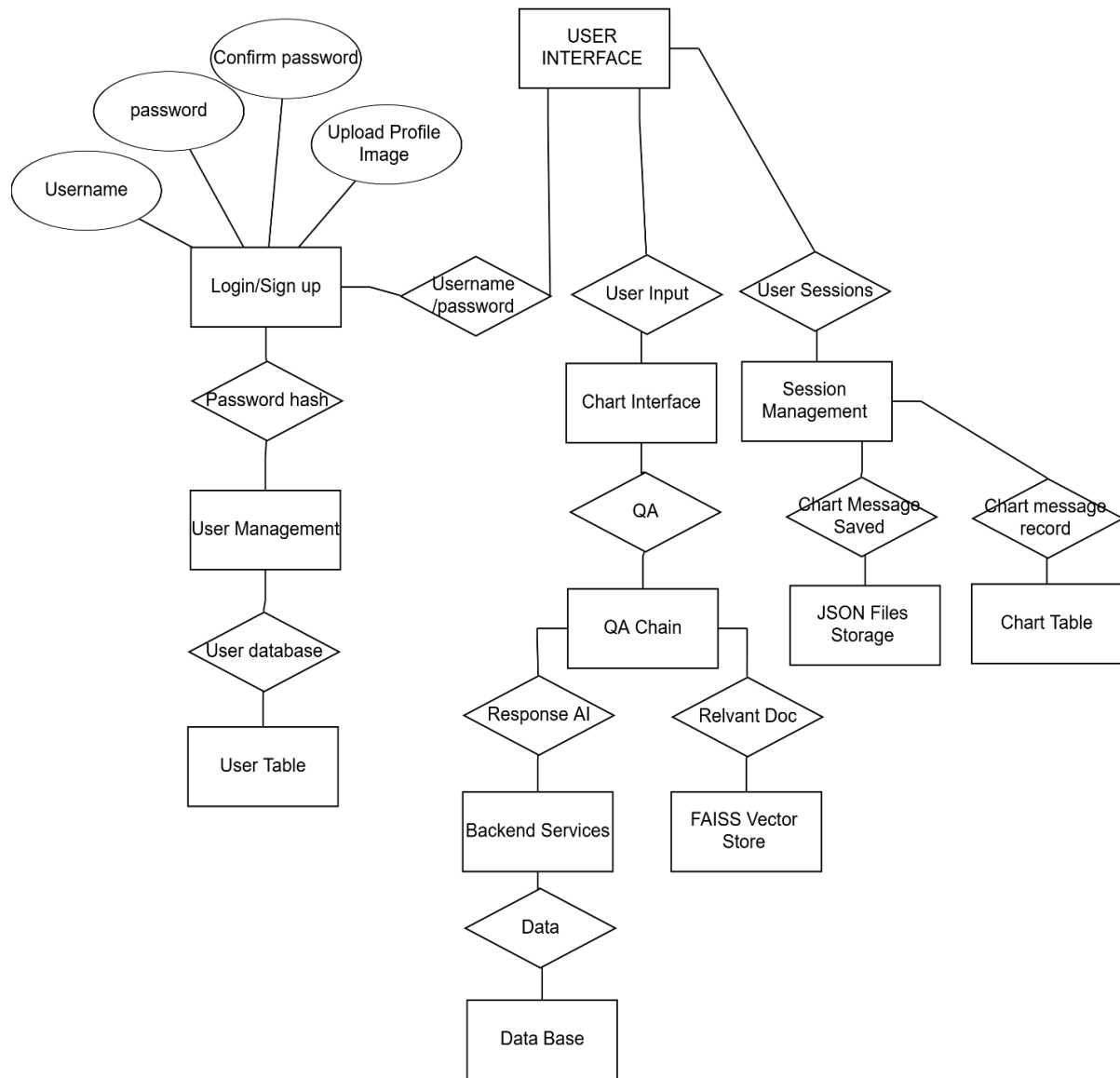


Figure 2.2 ERD Diagram

PSUEDOCODE

```
// Load necessary libraries and configurations
LOAD libraries and configuration files LOAD
environment variables

// Initialize and configure database

CREATE and configure database with user and chat history tables


// Function to hash passwords for security
FUNCTION hash_password(password):
    RETURN SHA256 hash of password


// Function to get AI model chain for question answering FUNCTION
get_conversational_chain():
    DEFINE prompt_template with context and question placeholders
    INITIALIZE AI model with settings
    INITIALIZE prompt using prompt_template
    LOAD QA chain with model and prompt
    RETURN chain


// Function to process user input and generate response FUNCTION
user_input(user_question):
    INITIALIZE embeddings using GoogleGenerativeAIEmbeddings
    LOAD FAISS vector store with embeddings
    PERFORM similarity search on user_question
    GET conversational chain
    EXECUTE chain with input documents and question RETURN
    generated response
```

```

FUNCTION display_typing_effect(response):
    INITIALIZE output as empty string CREATE
    a response container in UI
    FOR EACH character in response:
        APPEND character to output
        UPDATE response container with output

        SLEEP for a short duration to simulate typing effect RETURN
    output

// Function to save user session

FUNCTION save_session(username, session_name, session_data):
    GENERATE session_id using username and session_name SAVE
    session_data to file with session_id
    RETURN session_id

// Function to load a saved user session
FUNCTION load_session(session_id):
    TRY:

        OPEN session file with session_id
        LOAD session_data from file
        RETURN session_data
    EXCEPT FileNotFoundError:
        RETURN None

// Function to list all sessions for a user
FUNCTION list_sessions(username):
    INITIALIZE empty sessions list FOR
    EACH file in sessions directory:

```

```

        EXTRACT session_name from file name
        ADD session_name to sessions list
    RETURN sessions

// Function to delete a specific session
FUNCTION delete_session(session_id):
    TRY:

        DELETE session file with session_id
    EXCEPT FileNotFoundError:
        PASS

// Function to delete all sessions for a user
FUNCTION delete_all_sessions(username):
    FOR EACH file in sessions directory: IF
        file starts with username:
            DELETE file

// Main application logic
FUNCTION main():
    IF sessions directory does not exist: CREATE
        sessions directory

// Custom UI styling and layout

CONFIGURE UI layout with headers, buttons, and sidebar

// Check authentication status IF
authentication_status is True:
    DISPLAY user profile and logout button
    ALLOW user to start a new session

```

```

// Chat interface

IF "messages" not in session_state: INITIALIZE
    messages with assistant greeting

DISPLAY messages in chat format

GET user input from chat interface IF
input is provided:
    PROCESS input and generate response

    UPDATE messages with user and assistant responses SAVE
    session
ELSE:

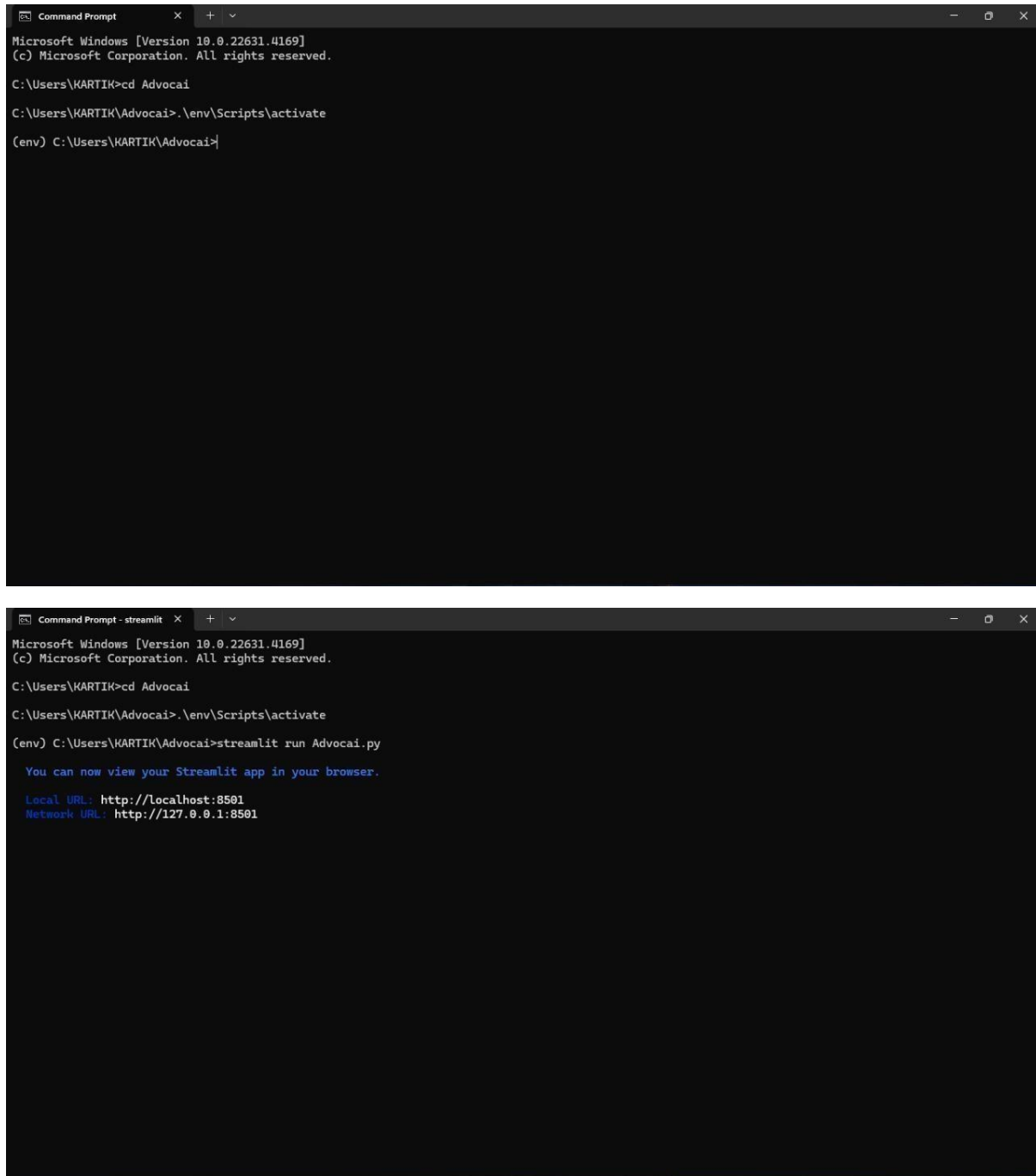
    DISPLAY login and signup options IF
    user chooses Login:
        GET username and password input
        VERIFY credentials with database IF
        credentials are valid:
            UPDATE authentication status and user profile in session_state
            RELOAD interface
        ELSE:

            DISPLAY error message
    ELSE IF user chooses Sign Up:
        GET new username, password, and profile image VERIFY
        password match
        SAVE new user to database DISPLAY
        success or error message

```

EXPERIMENTS AND RESULTS

Environment setup:



```
Command Prompt
Microsoft Windows [Version 10.0.22631.4169]
(c) Microsoft Corporation. All rights reserved.

C:\Users\KARTIK>cd Advocai
C:\Users\KARTIK\Advocai>.env\Scripts\activate
(env) C:\Users\KARTIK\Advocai>

Command Prompt - streamlit
Microsoft Windows [Version 10.0.22631.4169]
(c) Microsoft Corporation. All rights reserved.

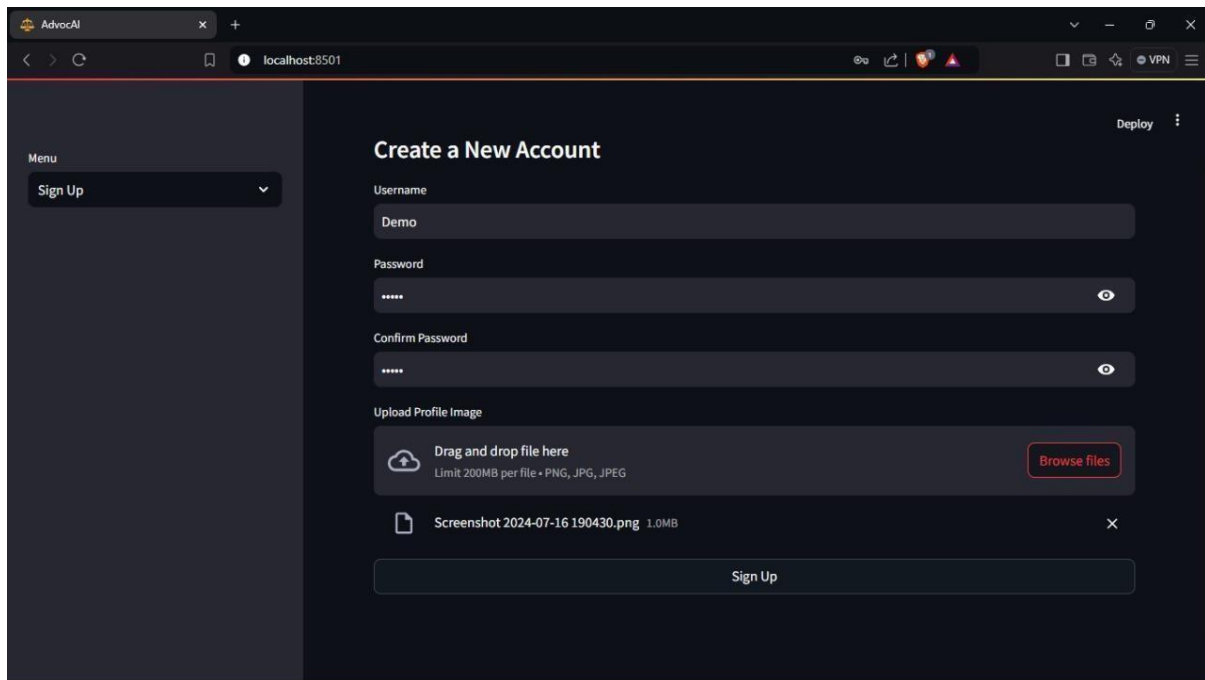
C:\Users\KARTIK>cd Advocai
C:\Users\KARTIK\Advocai>.env\Scripts\activate
(env) C:\Users\KARTIK\Advocai>streamlit run Advocai.py

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://127.0.0.1:8501
```

Figure 2.3 Environment Setup

Register new user:

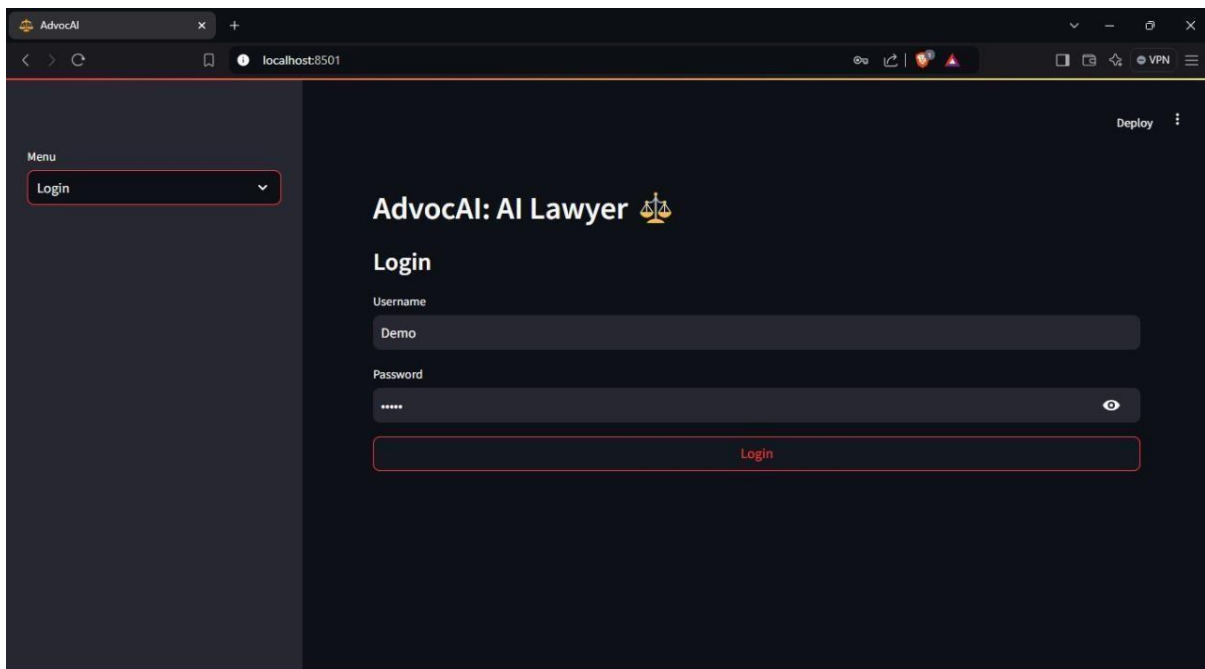


The screenshot shows a web browser window with the URL `localhost:8501`. The application is titled "AdvocAI". On the left, a "Menu" dropdown is set to "Sign Up". The main content area is titled "Create a New Account" and includes a "Deploy" link in the top right. The form contains the following fields and elements:

- Username:** A text input field containing the value "Demo".
- Password:** A password input field with masked characters "*****" and a toggle icon.
- Confirm Password:** A password input field with masked characters "*****" and a toggle icon.
- Upload Profile Image:** A section with a cloud icon and the text "Drag and drop file here" and "Limit 200MB per file • PNG, JPG, JPEG". A "Browse files" button is on the right.
- File Upload:** A list showing a file named "Screenshot 2024-07-16 190430.png" with a size of "1.0MB" and a close icon.
- Sign Up:** A large button at the bottom of the form.

Figure 2.4 Register New User

Login Dashboard:



The screenshot shows the same web browser window with the URL `localhost:8501`. The application is titled "AdvocAI". On the left, the "Menu" dropdown is now set to "Login". The main content area is titled "AdvocAI: AI Lawyer" with a scales of justice icon, followed by a "Login" sub-header. The form contains the following fields and elements:

- Username:** A text input field containing the value "Demo".
- Password:** A password input field with masked characters "*****" and a toggle icon.
- Login:** A large button at the bottom of the form.

Figure 2.5 Login Dashboard

Homepage:

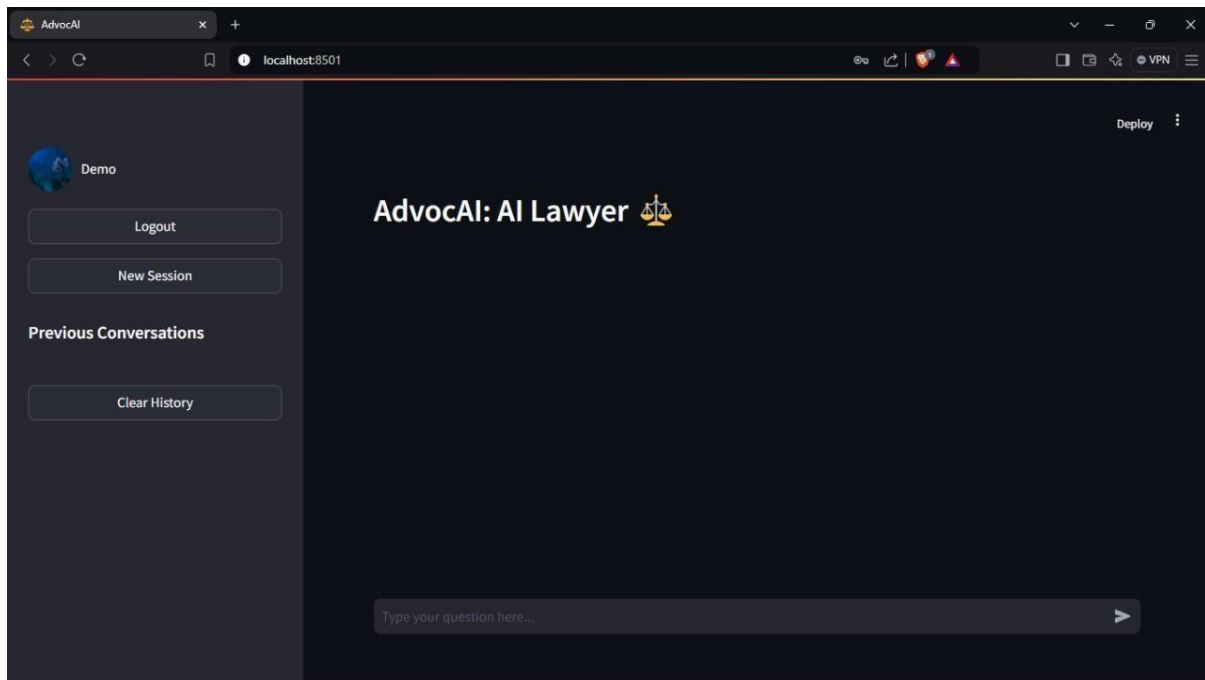


Figure 2.6 Homepage

Prompt:

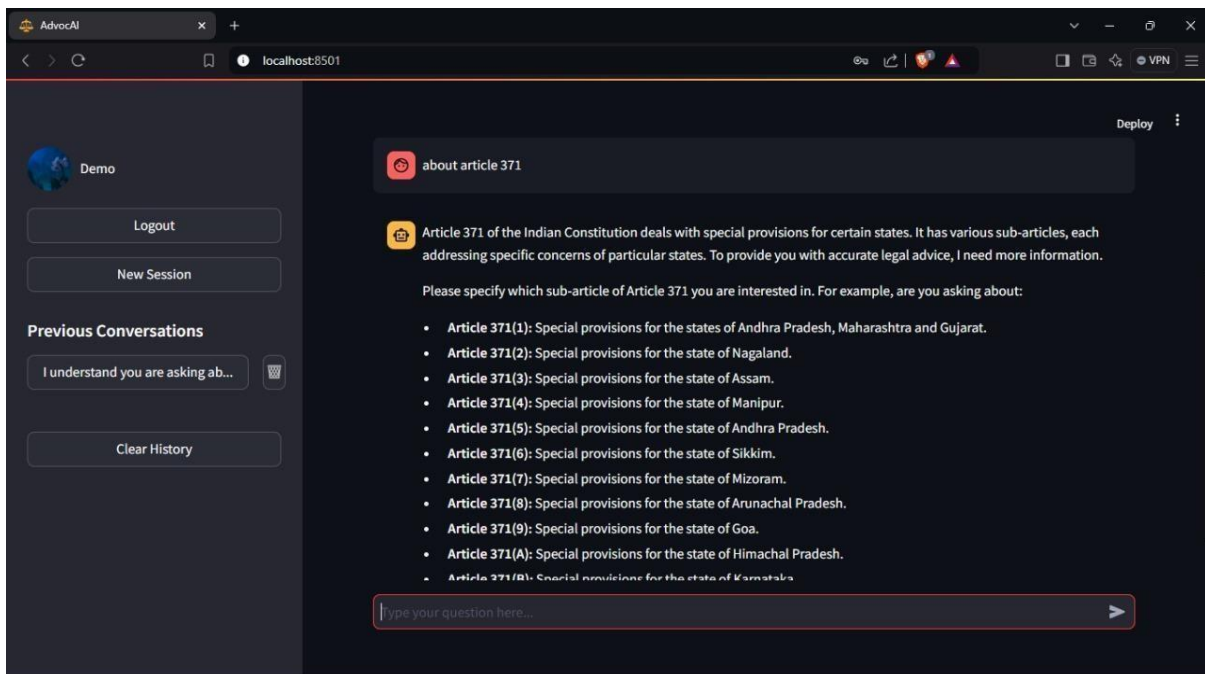


Figure 2.7 Prompt

TEST-CASES

Test Case 1: Login Button

Test Case ID: 001

Description: Verify that clicking the "Login" button redirects the user to the login page.

Preconditions:

- [1] User is on the homepage of the website or application.
- [2] "Login" button is visible on the navigation bar or a designated section.

Steps:

- [1] Navigate to the homepage.
- [2] Click on the "Login" button in the header/navigation bar.

Expected Result:

- [1] The user should be redirected to the login page, where they can input their credentials (username and password).

Actual Result:

- [1] To be filled after executing the test. For example: "User is successfully redirected to the login page" or "Login button is non-functional, no redirection happens."

Test Case 2: Signup Button

Test Case ID: 002

Description: Verify that clicking the "Sign Up" button opens the sign-up form and that the form allows input for username, password, and photo upload.

Preconditions:

- [1] User is on the homepage of the website or application.
- [2] "Sign Up" button is visible on the navigation bar or a designated section.

Steps:

- [1] Navigate to the homepage.
- [2] Click on the "Sign Up" button in the header/navigation bar.
- [3] Fill in the Username field with valid input.
- [4] Fill in the Password field with a valid password.
- [5] Upload a valid photo in the Photo Upload field.
- [6] Click the "Submit" button.

Expected Result:

- [1] The SignUp form should appear with fields for Username, Password, and Photo Upload.
- [2] User should be able to input text into the Username and Password fields.
- [3] User should be able to upload a valid photo file (e.g., JPEG, PNG).
- [4] After clicking "Submit," the form should validate the inputs and create a new account, redirecting the user to a confirmation page or the dashboard.

Actual Result:

- [1] To be filled after executing the test. For example: "Sign Up form appears, user can input data and upload a photo, and account is successfully created" or "Form fails to upload photo or validate inputs, showing errors."

Test Case 3: New Session

Test Case ID: 003

Description: Verify that clicking the "New Session" button starts a new session automatically without requiring any additional input.

Preconditions:

- [1] User is logged into the website or application.
- [2] "New Session" button is visible and enabled on the page.

Steps:

- [1] Navigate to the page where the "New Session" button is located.
- [2] Click on the "New Session" button.

Expected Result:

- [1] A new session should start immediately upon clicking the button.
- [2] The user should be redirected to the new session page, where the session details such as session ID or status are displayed.
- [3] No further input or confirmation should be required from the user.

Actual Result:

- [1] To be filled after executing the test. For example: "New session started successfully, and user is redirected to the session page" or "New session button is non-functional, no session is started."

Test Case 4: Clear History

Test Case ID: 004

Description: Verify that clicking the "Clear History" button prompts the user for confirmation before clearing the user's history.

Preconditions:

- [1] User is logged into the website or application.
- [2] User has some history recorded (e.g., previous sessions, actions).
- [3] "Clear History" button is visible and enabled on the page.

Steps:

- [1] Navigate to the page where the "Clear History" button is located (e.g., settings or history section).
- [2] Click on the "Clear History" button.
- [3] A confirmation dialog should appear asking, "Are you sure you want to clear the history?" with options like "Yes" and "No".
- [4] Click "Yes" to confirm.

Expected Result:

- [1] After clicking "Yes", the user's history (e.g., session data, search history) should be cleared immediately.
- [2] A confirmation message (e.g., "History Cleared Successfully") should be displayed.
- [3] If "No" is clicked, the action should be canceled, and the history should remain unchanged.

Actual Result:

- [1] To be filled after executing the test. For example: "History cleared successfully after confirmation" or "Clear History button is non-functional, no confirmation dialog appears."

Test Case 5: Prompt

Test Case ID: 005

Description: Verify that clicking the "Prompt" button accepts user input for legal advice and provides a valid response, while filtering out foul language (including adult language) and hate speech without displaying output for such inputs.

Preconditions:

- [1] User is logged into the website or application.
- [2] "Prompt" panel is visible and enabled on the page.

Steps:

- [1] Navigate to the page where the "Prompt" panel is located.
- [2] Click on the "Prompt" panel.
- [3] A prompt input dialog should appear, asking the user for input (e.g., "Type Your question Here:").
- [4] Enter a valid legal question (e.g., "What are the rights of a tenant?") and click the right arrow icon to submit.

Expected Result:

- [1] For the valid legal question, the application should process the input and provide an appropriate legal response (e.g., "Tenants have the right to a safe living environment.").
- [2] For inputs containing foul language or adult content, the application should not display any output and should provide a message indicating that such language is not acceptable (e.g., "Please avoid using inappropriate language.").

Actual Result:

- [1] To be filled after executing the test. For example: "Legal question processed successfully, foul language and hate speech filtered out as expected" or "Prompt button is non-functional, or incorrect responses were given."

FUTURE SCOPE

The scope of the Advoc AI project encompasses a comprehensive range of legal, technological, and user-centric elements aimed at developing an AI-driven legal advisory platform tailored to the Indian legal system. It includes robust legal framework coverage across various domains such as civil, criminal, corporate, family, and property law. The technological foundation of the platform is built upon advanced natural language processing and machine learning algorithms, enabling the AI to interpret legal language accurately and improve the quality of legal advice over time. Additionally, the system will integrate an extensive legal database containing statutes, case laws, and regulations to support informed responses.

On the user experience and platform delivery front, Advoc AI is designed to offer an intuitive interface accessible across multiple devices, supported by scalable cloud infrastructure for optimal performance. Key aspects include user training resources, comprehensive testing for functionality and usability, and stringent validation of legal outputs against expert opinions. The project also prioritizes data security and compliance with legal standards, ensuring user confidentiality and regulatory adherence. Post-deployment, the platform will undergo continuous maintenance and updates to reflect the latest legal developments and technological advancements, guaranteeing reliability and relevance for users.

CONCLUSION

The AdvocAI is an innovative initiative that harnesses the power of artificial intelligence to make legal advice more accessible and affordable, especially for individuals in remote or underserved areas. By offering digital legal support, AdvocAI aims to bridge the gap between professional legal services and those who may not have easy access to them due to geographical or financial limitations.

Central to AdvocAI's functionality is the integration of natural language processing (NLP) and machine learning, which allow the platform to interpret and respond to complex legal queries in a simplified manner. These technologies help overcome traditional challenges in the legal domain, including the intricate nature of legal language, the importance of model transparency, and the necessity of maintaining fairness and impartiality in automated legal assistance.

REFERENCES

Links:

- [1] <https://indiaai.gov.in/news/lawyerdesk-advocacy-unveiled-advocase-ai-an-ai-assistant-for-legal-professionals-in-india>
- [2] <https://www.expresscomputer.in/news/advocase-ai-unveiled-a-revolutionary-ai-assistant-for-legal-professionals-in-india>
- [3] <https://indiaadvocacy.in>

Papers:

- [1] IEEE Xplore Digital Library: Access a vast collection of research papers and articles across various engineering and technology disciplines.
- [2] IEEE Xplore Author Guidelines for Integrative Literature Reviews: Offers detailed instructions on formatting and organizing literature reviews. procomm.ieee.org

CHAPTER-3

ON JOB TRAINING – 2 “ONLINE GROCERY STORE”

ABSTRACT

The Online Grocery Store is a modern, web-based eCommerce application designed to simplify the process of purchasing groceries online. It offers a user-friendly interface that enables customers to browse a wide variety of grocery items, view detailed product descriptions, and effortlessly add items to their shopping cart. With a focus on ease of use and accessibility, the platform aims to provide a seamless and efficient shopping experience for users of all backgrounds.

At the heart of the application is a robust system that supports user account creation and authentication. Once registered, users can log in to access personalized dashboards where they can update their personal details, manage saved addresses, and view past orders. This personalized experience helps users track their shopping history and facilitates faster checkouts in future transactions, enhancing the overall convenience of the platform.

The platform incorporates essential eCommerce functionalities such as dynamic product search, real-time cart management, and a streamlined checkout process. It also supports various payment options, including the integration of a secure digital wallet system, providing users with flexible and safe transaction methods. These features work together to ensure that users have full control over their shopping journey, from product selection to final payment.

Built using modern web technologies such as HTML, CSS, JavaScript, Bootstrap, Node.js, and MongoDB, the Online Grocery Store is both scalable and responsive. The back-end architecture ensures fast data handling and storage, while the front-end delivers an intuitive and responsive design. Together, these technologies create a reliable, interactive, and performance-optimized application that addresses the growing demand for convenient online grocery shopping.

INTRODUCTION

The Online Grocery Store project aims to simplify the process of purchasing groceries online by providing a user-friendly and accessible platform. With the growing popularity of online shopping, this application offers a convenient way for users to browse, search, and buy grocery items from the comfort of their homes. It features a responsive design, secure login system, and smooth navigation to enhance the overall user experience.

In addition, the system is built with strong backend support to manage data efficiently and ensure reliable performance. Secure transactions, real-time updates, and easy-to-use features make the platform suitable for users of all ages. As eCommerce continues to evolve, this project serves as a practical solution for modern grocery shopping needs.

The Online Grocery Store project is designed to simplify and enhance the grocery shopping experience by offering a convenient digital platform. With the growing demand for online services, this application allows users to browse, search, and purchase grocery items from the comfort of their homes. It features a responsive and intuitive interface, secure user authentication, and efficient database management to ensure a smooth and user-friendly experience.

As eCommerce continues to reshape the way people shop, this project embraces modern web technologies to provide a reliable and secure solution. The system prioritizes seamless navigation, fast performance, and secure transactions, supported by a robust backend that efficiently handles user and product data. Overall, the Online Grocery Store contributes to the digital transformation of everyday shopping.

EXISTING SYSTEM AND LITERATURE SURVEY

Existing System

Traditional grocery shopping involves visiting physical stores, which is time-consuming and lacks personalized recommendations. Existing online grocery platforms offer convenience but may suffer from limited payment options, inefficient search mechanisms, and inadequate user account management.

The existing systems for online grocery shopping often lack user-friendly interfaces, efficient search functionalities, and robust payment gateways. Many platforms also face challenges in managing user data securely and providing real-time updates on product availability.

Literature Survey

Several studies highlight the increasing adoption of online grocery shopping. Key factors influencing user preferences include ease of navigation, payment security, product availability, and personalized recommendations. Our system incorporates these insights to enhance user satisfaction.

A literature survey was conducted to understand the current trends and technologies in eCommerce and online grocery shopping. The survey highlighted the importance of responsive design, secure authentication, and efficient database management in enhancing user experience and ensuring data security.

PROBLEM STATEMENT AND OBJECTIVES

Problem Definition/Aim of the Project

The primary problem addressed by this project is the lack of a user-friendly and efficient online grocery shopping platform that provides a seamless shopping experience, secure transactions, and real-time product updates.

Develop a user-friendly, secure, and efficient online grocery shopping platform that allows users to browse, search, purchase, and manage their orders.

Challenges

- [1] Ensuring secure user authentication and data protection.
- [2] Providing an efficient search and filtering mechanism for products.
- [3] Managing real-time updates on product availability and pricing.
- [4] Ensuring a smooth user experience with a well-structured UI.
- [5] Implementing secure authentication and payment processing.

Limitations

- [1] The project is a sample implementation and may not cover all real-world scenarios.
- [2] Limited scalability due to the use of a local database.
- [3] Requires internet connectivity.
- [4] Limited to registered users for making purchases.
- [5] Digital wallet functionality has transaction limits.

Applications

- [1] Online grocery shopping for individual users.
- [2] Management of user accounts, orders, and transactions.
- [3] Real-time product search and filtering.
- [4] Online grocery shopping.
- [5] Secure digital transactions.

REQUIREMENTS

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4GB or more
- Storage: 10GB free disk space
- Operating System: Windows/Linux/MacOS
- A computer with a minimum of 4GB RAM and a dual-core processor.
- Internet connection for database access and deployment.

Software Requirements

- **Frontend Languages:** HTML, CSS, React and Client-Side JavaScript Frontend Framework: Bootstrap Frontend Library: jQuery
- **Backend:** Server-Side JavaScript Backend Framework: NodeJS and NPM Database: MongoDB Other Libraries: Tota11y, and HTML Validator Tools: MS Visual Code, MS Command Panel

Functional Requirements

- User authentication (Sign Up / Sign In)
- Product search and filtering
- Cart management
- Order processing and history tracking
- Digital wallet for payments
- User registration and login.
- Product search and filtering.
- Cart management and checkout process.
- User dashboard for managing account settings, orders, and payments

Non-Functional Requirements

- Scalability to support multiple users
- Secure transactions
- User-friendly interface
- Responsive and user-friendly interface.
- Secure authentication and data protection.
- Efficient database management and real-time updates.

SYSTEM DESIGN AND METHODOLOGY

Methodology

The project follows an iterative development approach, with phases including requirement gathering, design, development, testing, and deployment.

Software Architecture

The system follows a client-server model, with a frontend built using HTML, CSS, JavaScript, Bootstrap, and jQuery, and a backend using Node.js, Express, and MongoDB.

Database Design

The database consists of collections for users, products, orders, and transactions. MongoDB is used for data storage.

UI Design

The interface includes a homepage with product listings, a search bar, a cart view, and a user dashboard.

Module Design

User Management:

- Registration, authentication, profile management.

Product Management:

- Search, filter, and categorize products.

Cart Management:

- Add, remove, and modify cart items.

Order Processing:

- Checkout and payment processing

ERD DIAGRAM

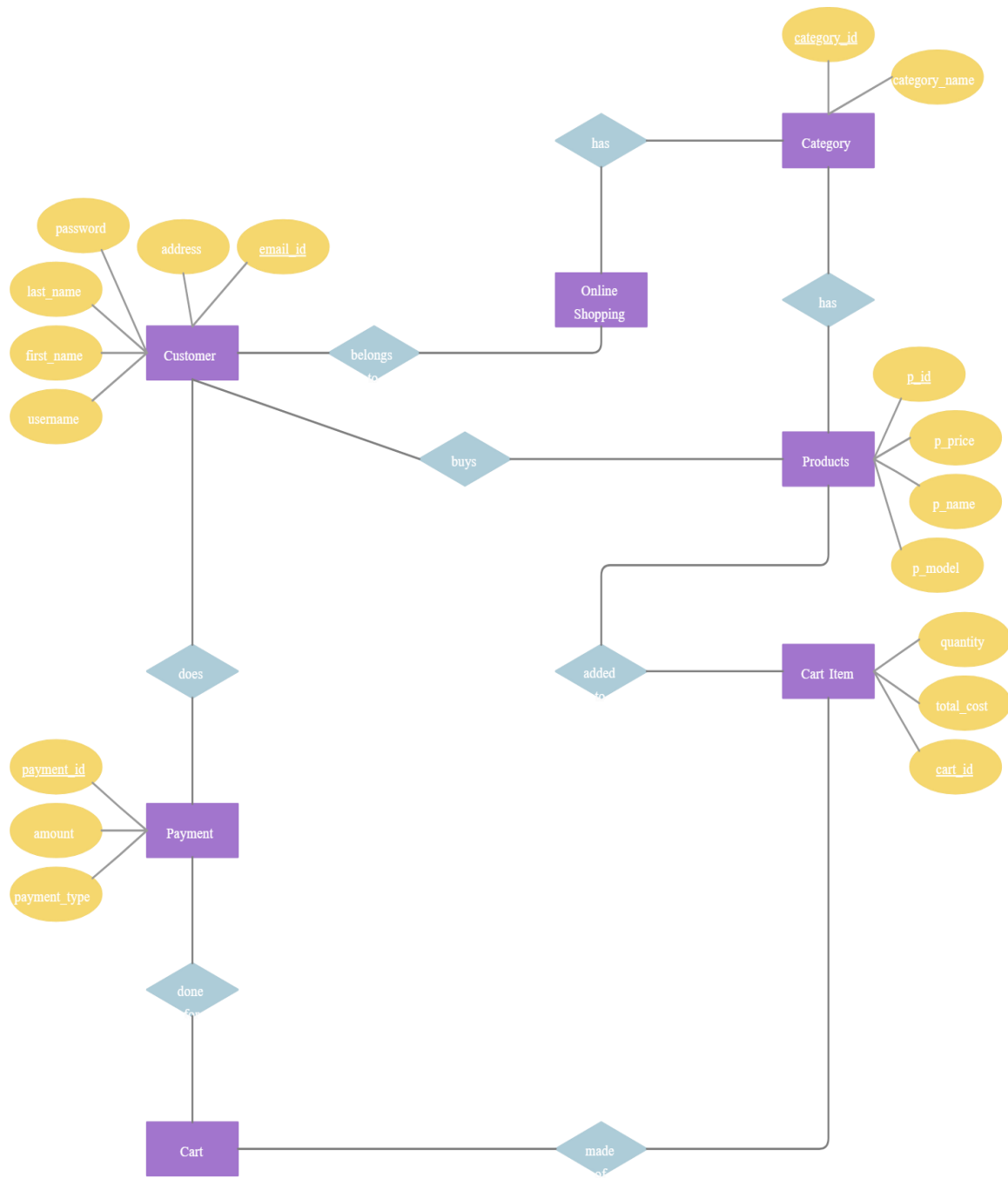


Figure 3.1 ERD Diagram

EXPERIMENTS AND RESULTS

Environment Setup:

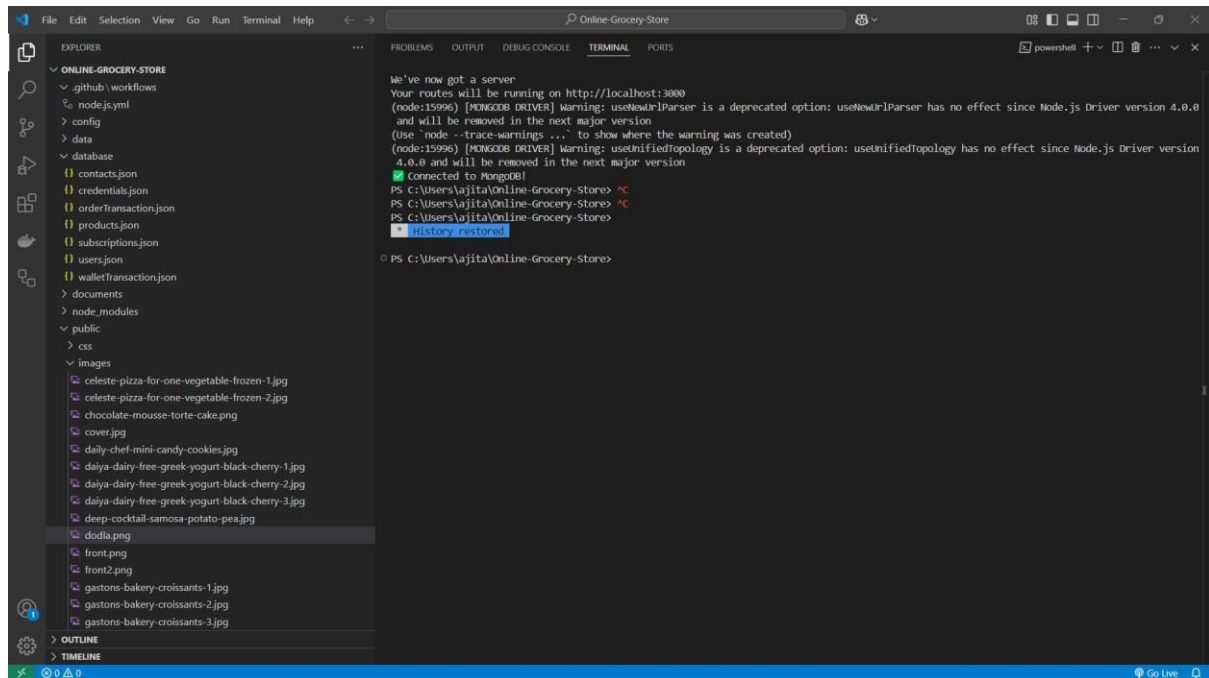


Figure 3.2 Environment Setup

Home Page:

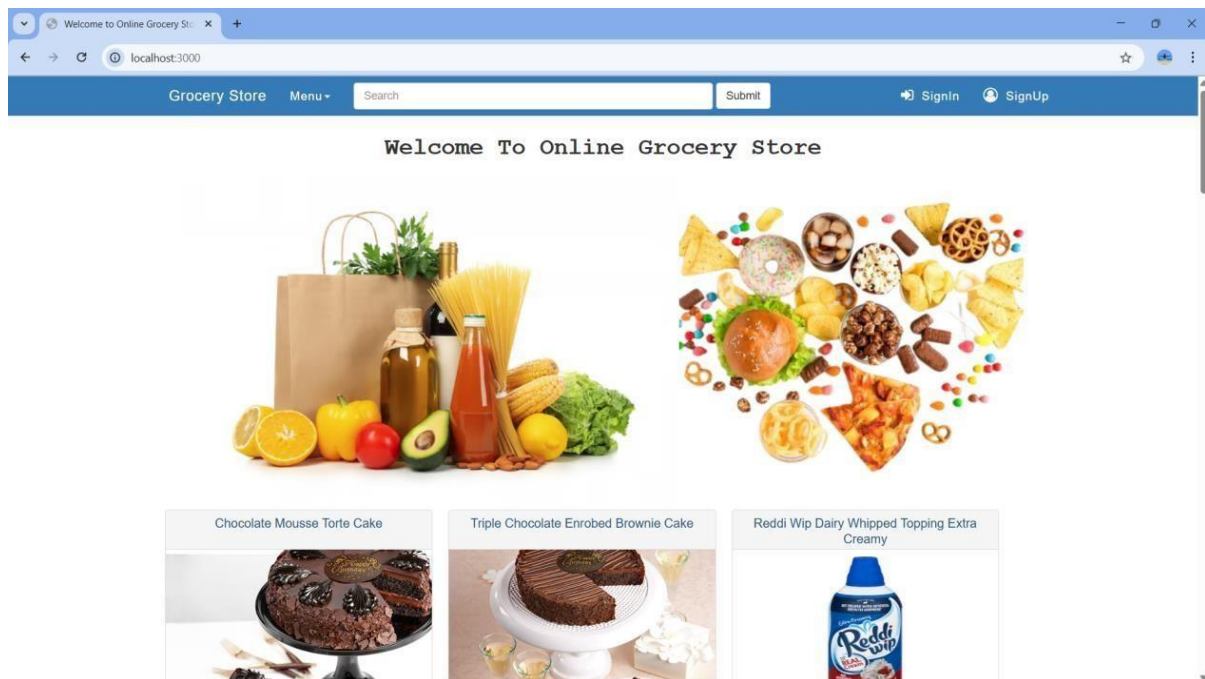


Figure 3.3 Home Page

Register New User:

The screenshot shows a web browser window with the address bar displaying 'localhost:3000/user/new'. The page has a blue header with 'Grocery Store', a 'Menu' dropdown, a search bar, and 'Signin' and 'SignUp' links. The main content area features a 'Create a new Account' form with the following fields: Name (filled with 'Second Project'), Email Id (filled with 'ajitalagundics@gmail.com'), Contact Number (filled with '8197140817'), Password (masked with '****'), and Confirm Password (masked with '****'). A 'Create Account' button is at the bottom of the form. Below the form is a dark blue footer with three sections: 'About Project' (describing the eCommerce application), 'Contact Us' (with fields for Full Name, Email Id, Contact Number, and Description, and a 'Send' button), and 'Follow Us' (with a GitHub link and a 'Subscribe' button with an email input field).

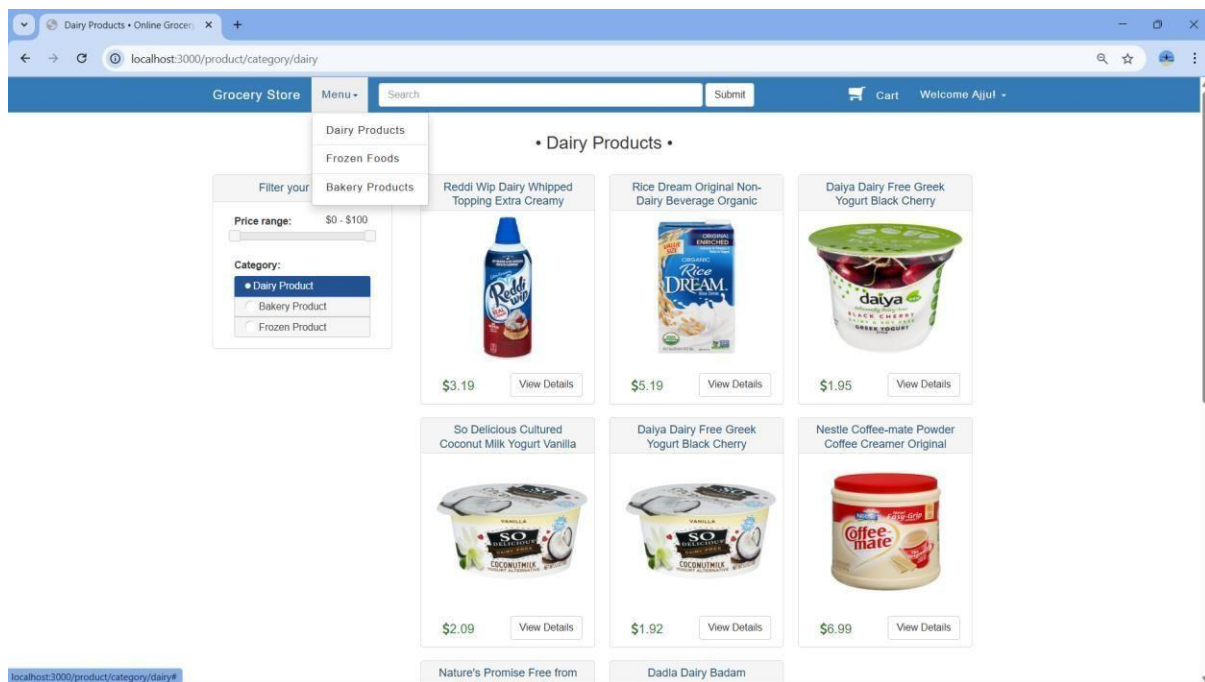
Figure 3.4 Register New User

Login Dashboard:

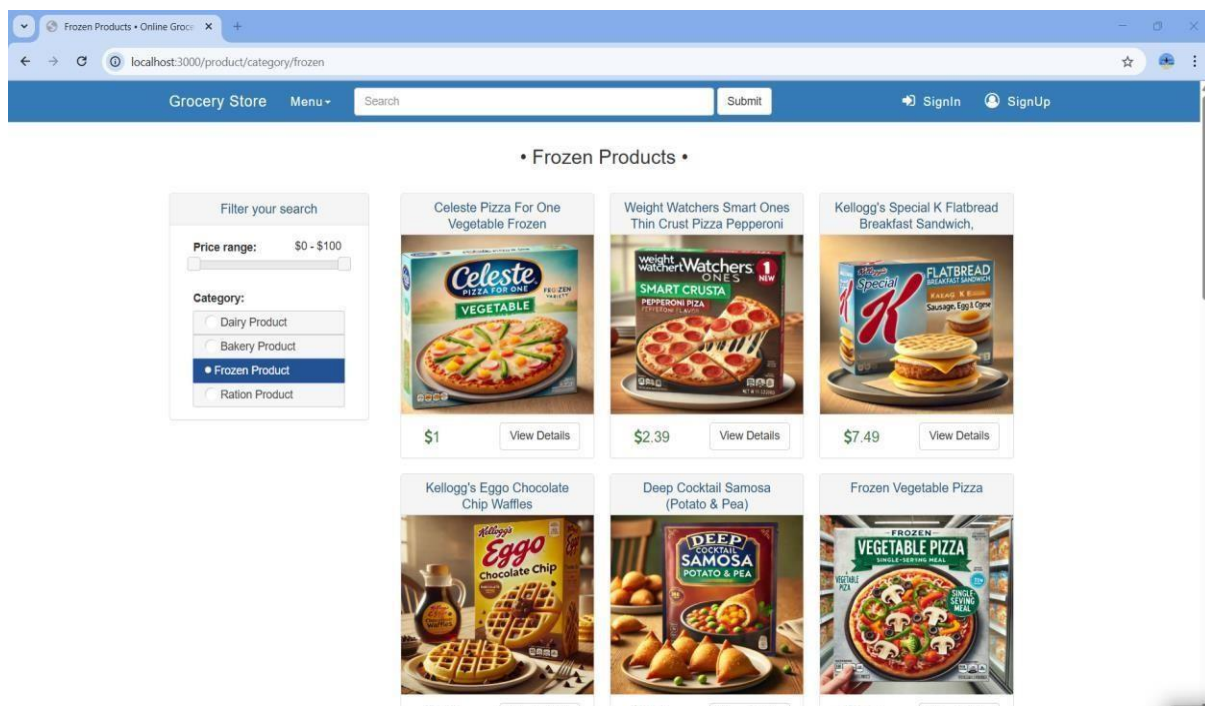
The screenshot shows a web browser window with the address bar displaying 'localhost:3000/user/login'. The page has a blue header with 'Grocery Store', a 'Menu' dropdown, a search bar, and 'Signin' and 'SignUp' links. The main content area features a 'User Dashboard Login' form with the following fields: Email Id (filled with 'ajitalagundics1@gmail.com') and Password (masked with '****'). There are 'Login' and 'Forgot Password' buttons at the bottom of the form. Below the form is a dark blue footer with three sections: 'About Project' (describing the eCommerce application), 'Contact Us' (with fields for Full Name, Email Id, Contact Number, and Description, and a 'Send' button), and 'Follow Us' (with a GitHub link and a 'Subscribe' button with an email input field).

Figure 3.5 Login Dashboard

Dairy Product Home Page:



Frozen Product Home Page:



Bakery Product Home Page:

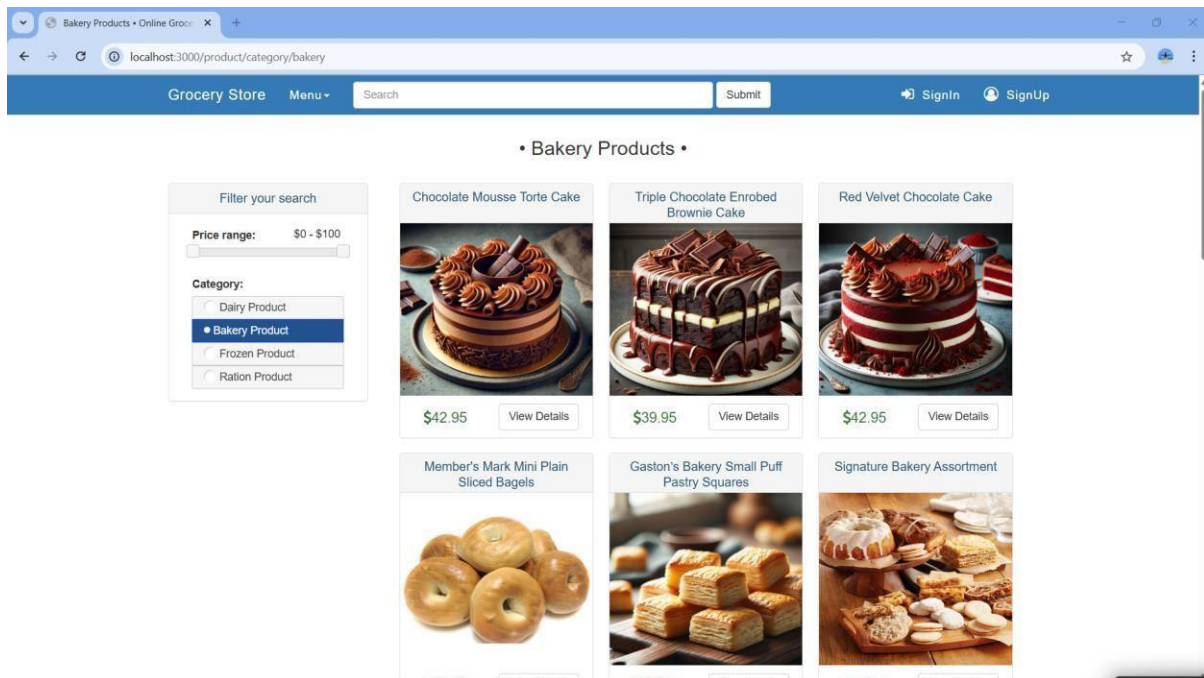


Figure 3.6 Product Home Pages

Selecting Product:

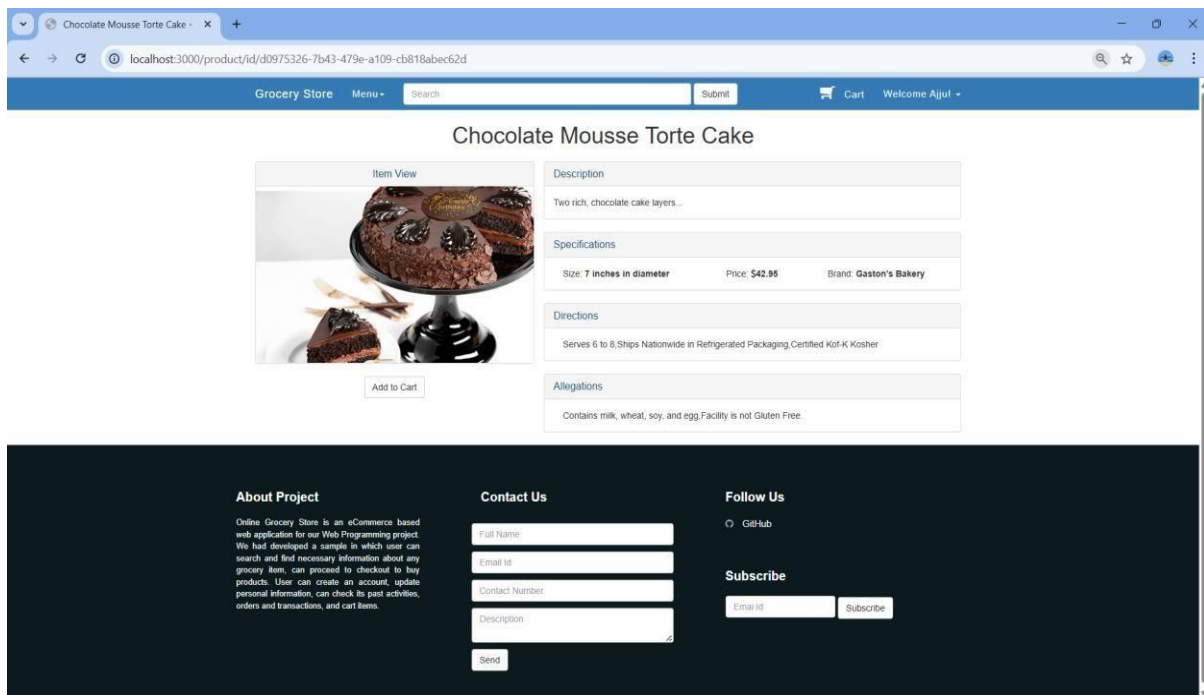


Figure 3.7 Selecting Product

Adding To Cart:

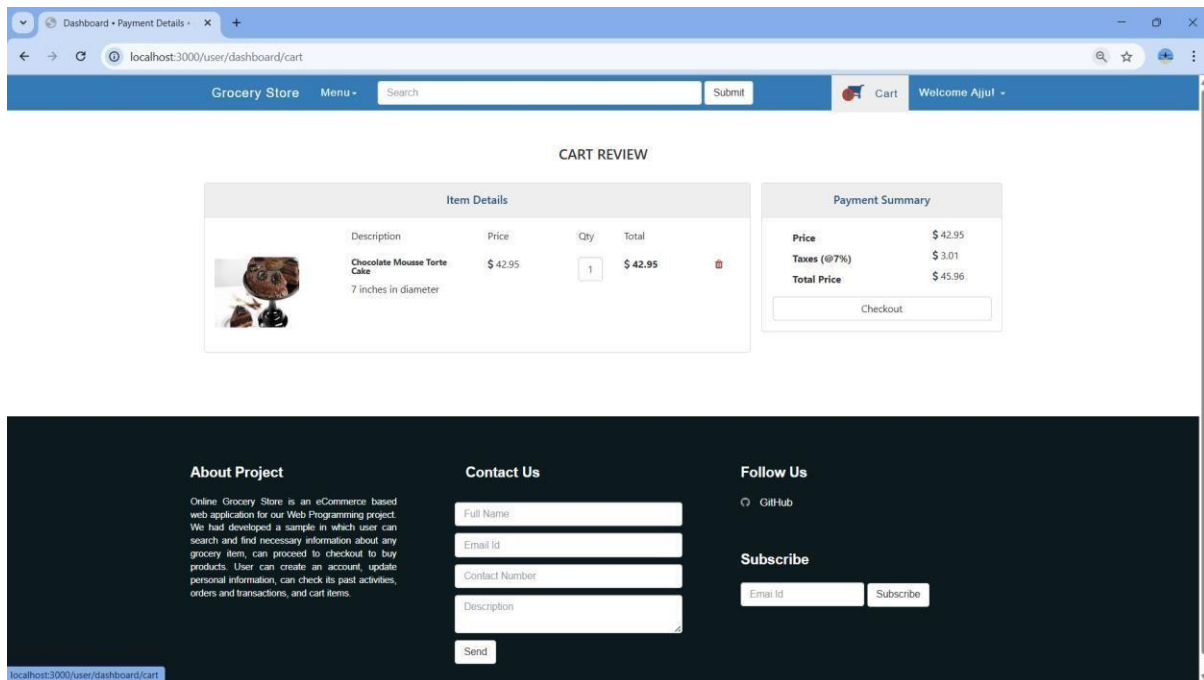


Figure 3.8 Adding To Cart

Payment Of The Product:

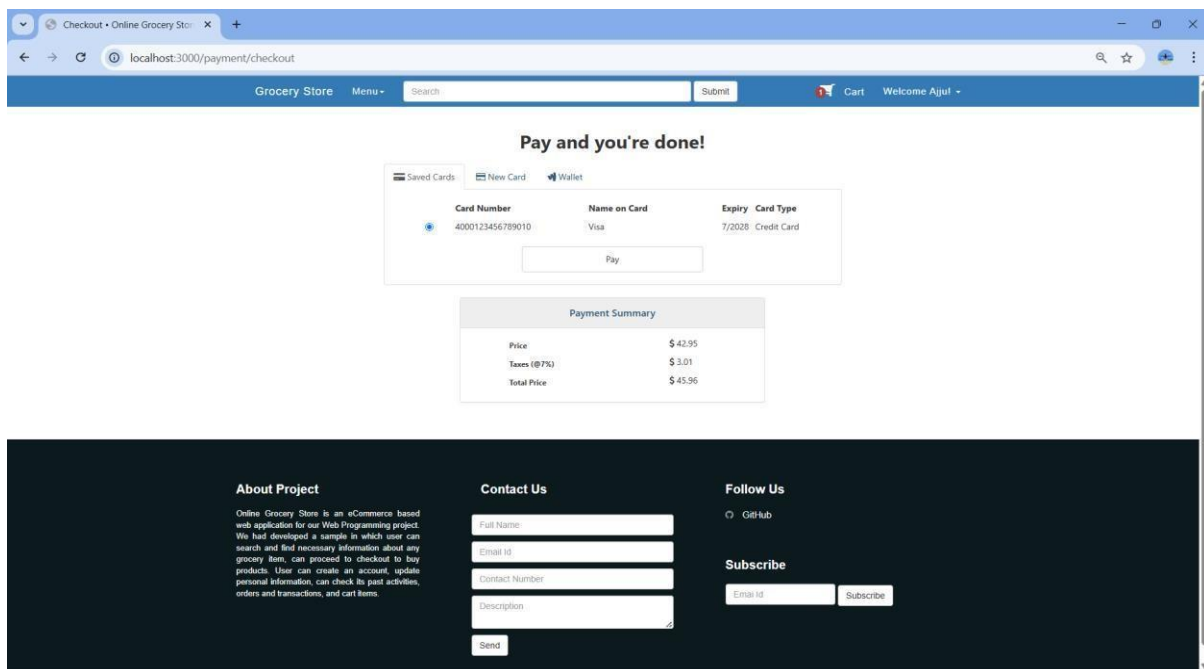


Figure 3.9 Payment of the Product

Order Has Been Successful:

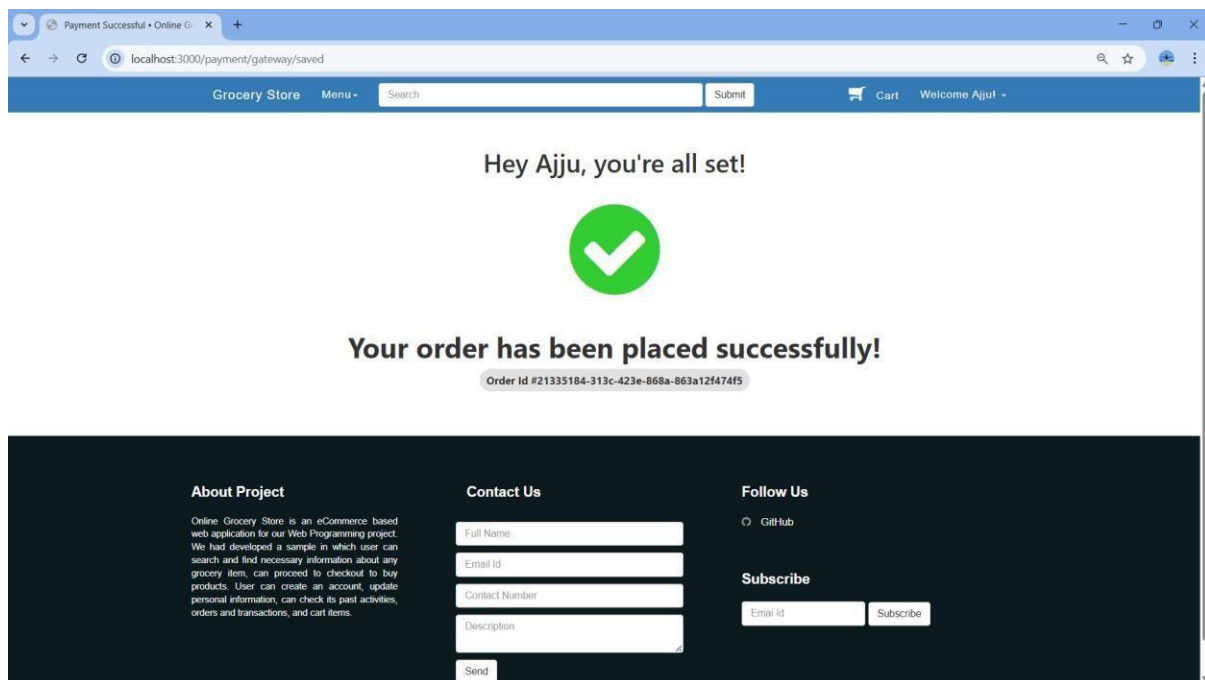


Figure 3.10 Order Has Been Successful

Order History:

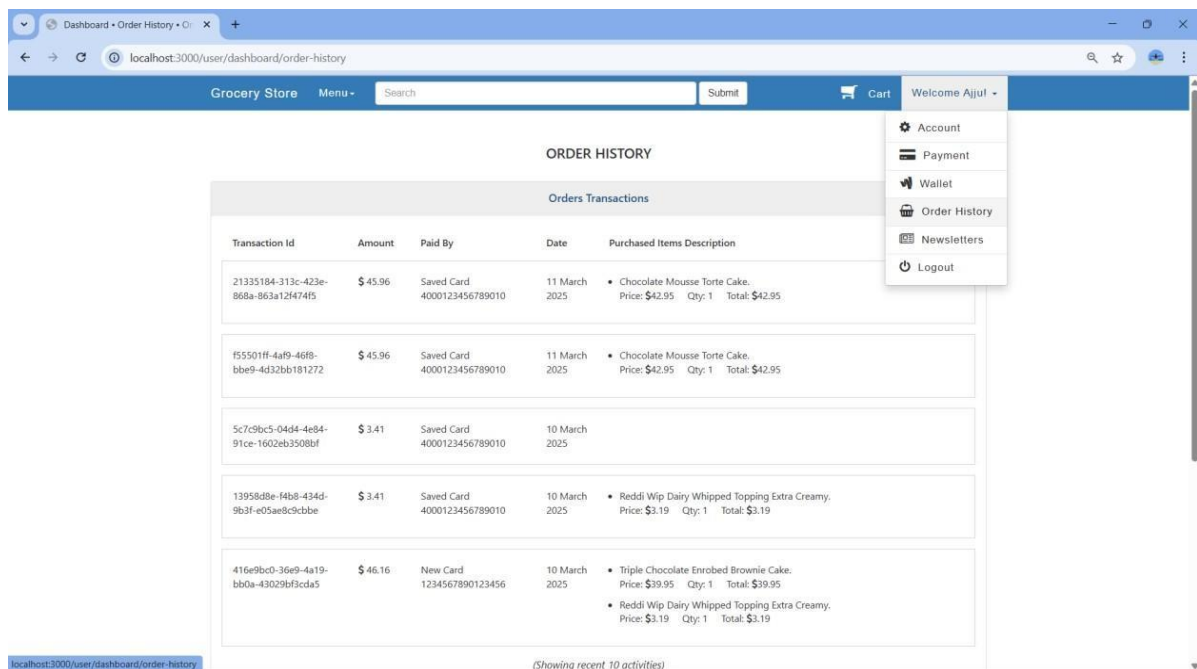


Figure 3.11 Order History

Wallet Information:

The screenshot shows a web application interface for 'Wallet Information'. The browser address bar indicates the URL is `localhost:3000/user/dashboard/wallet`. The page features a blue header with 'Grocery Store', 'Menu', a search bar, and a 'Submit' button. A user profile dropdown for 'Welcome Ajul' is open, showing options like 'Account', 'Payment', 'Wallet', 'Order History', 'Newsletters', and 'Logout'. The main content area is titled 'WALLET INFORMATION' and contains two sections: 'Available Cash' showing '\$ 0' and 'Wallet Transactions' showing 'No wallet transactions'. Below these is an 'Add Cash' form with fields for 'Enter Amount' (with a placeholder '\$ Enter amount to add'), 'Select Card' (with a dropdown 'Select card to pay'), and buttons for 'Use Saved Card' and 'Use New Card'. The footer is dark blue and includes 'About Project', 'Contact Us' (with fields for Full Name, Email Id, Contact Number, and Description), 'Follow Us' (with a GitHub link), and a 'Subscribe' form (with an Email Id field and a Subscribe button).

Figure 3.12 Wallet Information

Payment Setting:

The screenshot shows a web application interface for 'Payment Setting'. The browser address bar indicates the URL is `localhost:3000/user/dashboard/payments`. The page features a blue header with 'Grocery Store', 'Menu', a search bar, and a 'Submit' button. A user profile dropdown for 'Welcome Ajul' is open, showing options like 'Account', 'Payment', 'Wallet', 'Order History', 'Newsletters', and 'Logout'. The main content area is titled 'PAYMENT SETTINGS' and contains two sections: 'Saved Cards' and 'Add Card'. The 'Saved Cards' section displays a table with the following data:

Card Number	Card Holder	Card Expiry	Card Type
4000123456789010	Visa	7/2028	Credit Card

The 'Add Card' section contains form fields for 'Name On Card' (with a placeholder 'Card Holder's Name'), 'Card Number', 'Card Type' (with a dropdown), 'Card Expiry' (with separate fields for Month and Year), and 'CVV' (with a placeholder 'CVV Number'). An 'Add' button is located at the bottom of the 'Add Card' section.

Figure 3.13 Payment Setting

Account Setting:

The screenshot shows a web browser window with the URL `localhost:3000/user/dashboard/account`. The page has a blue header with "Grocery Store", "Menu", a search bar, and a "Submit" button. On the right, there's a "Cart" icon and a "Welcome Aju!" dropdown menu. The main content area is titled "ACCOUNT SETTINGS" and contains two sections: "Profile Information" and "Privacy Setting".

ACCOUNT SETTINGS

Profile Information

Email Id:

Name:

Contact Number:

Privacy Setting

New Password:

Confirm Password:

The "Welcome Aju!" dropdown menu is open, showing the following options: Account, Payment, Wallet, Order History, Newsletters, and Logout.

Figure 3.14 Account Setting

TEST-CASES

Test Case 1: Login Button

Test Case ID: 001 Description: Verify that clicking the "Login" button redirects the user to the login page.

Preconditions:

- [1] User is on the homepage of the website or application.
- [2] "Login" button is visible on the navigation bar or a designated section.

Steps:

- [1] Navigate to the homepage.
- [2] Click on the "Login" button in the header/navigation bar.

Expected Result:

- [1] The user should be redirected to the login page, where they can input their credentials (username and password).

Actual Result:

- [1] To be filled after executing the test. For example: "User is successfully redirected to the login page" or "Login button is non-functional, no redirection happens."

Test Case 2: Signup Button

Test Case ID: 002 Description: Verify that clicking the "Sign Up" button opens the sign-up form and that the form allows input for username, password, and other required fields.

Preconditions:

- [1] User is on the homepage of the website or application.
- [2] "Sign Up" button is visible on the navigation bar or a designated section.

Steps:

- [1] Navigate to the homepage.
- [2] Click on the "Sign Up" button in the header/navigation bar.
- [3] Fill in the Username field with valid input.
- [4] Fill in the Password field with a valid password.
- [5] Fill in other required fields (e.g., email, phone number).
- [6] Click the "Submit" button.

Expected Result:

- [1] The SignUp form should appear with fields for Username, Password, and other required information.
- [2] User should be able to input text into the Username, Password, and other fields.
- [3] After clicking "Submit," the form should validate the inputs and create a new account, redirecting the user to a confirmation page or the dashboard.

Actual Result:

- [1] To be filled after executing the test. For example: "Sign Up form appears, user can input data, and account is successfully created" or "Form fails to validate inputs, showing errors."

Test Case 3: Add to Cart Button

Test Case ID: 003 Description: Verify that clicking the "Add to Cart" button adds the selected product to the user's cart.

Preconditions:

- [1] User is logged into the website or application.
- [2] User is viewing a product details page.
- [3] "Add to Cart" button is visible and enabled on the page.

Steps:

- [1] Navigate to a product details page.
- [2] Click on the "Add to Cart" button.
- [3] Navigate to the cart page to verify the product has been added.

Expected Result:

- [1] The selected product should be added to the user's cart.
- [2] The cart page should display the added product with the correct quantity and price.

Actual Result:

- [1] To be filled after executing the test. For example: "Product successfully added to the cart" or "Add to Cart button is non-functional, product not added."

Test Case 4: Checkout Process

Test Case ID: 004 Description: Verify that the checkout process allows the user to review their order, select a payment method, and complete the purchase.

Preconditions:

- [1] User is logged into the website or application.
- [2] User has items in their cart.
- [3] "Checkout" button is visible and enabled on the cart page.

Steps:

- [1] Navigate to the cart page.
- [2] Click on the "Checkout" button.
- [3] Review the order summary.
- [4] Select a payment method (e.g., saved card, wallet, new card).
- [5] Click on the "Pay Now" button.

Expected Result:

- [1] The user should be able to review their order details.
- [2] The user should be able to select a payment method and complete the purchase.
- [3] A confirmation message should be displayed, and the order should be recorded in the user's order history.

Actual Result:

- [1] To be filled after executing the test. For example: "Checkout process completed successfully, order confirmed" or "Checkout button is non- functional, payment not processed."

Test Case 5: Search Functionality

Test Case ID: 005 Description: Verify that the search bar allows users to search for products by name, category, or brand and displays relevant results.

Preconditions:

- [1] User is on the homepage or any product listing page.
- [2] Search bar is visible and enabled on the page.

Steps:

- [1] Navigate to the homepage or product listing page.
- [2] Enter a product name, category, or brand in the search bar.
- [3] Press Enter or click the search icon.

Expected Result:

- [1] The search results page should display products matching the search query.
- [2] The results should be relevant to the entered search term (e.g., product name, category, or brand).

Actual Result:

- [1] To be filled after executing the test. For example: "Search results displayed correctly" or "Search functionality is non-functional, no results displayed."

Project Core Features

User Sign Up /Sign In: User need to create account to make any kind of purchases but to check projects and other searches does not require user to be logged in. First user need to be get registered and logged in to check for recent activities, transactions, access e-wallet and items cart. For creating account user need to click on signup button fulfill all the create an account form requirement to get registered with us. In case your wants to be logged in, he need to go to sign in and provide its credentials which has been set up with us. In case user has forgot its password, we provide an module to regenerate a new password. The link for forgot module it can be found in sign in.

[1] User Dashboard: To change user information and other settings we have developed modules in dashboard panel but user can also find all these options on the top right corner of the page when user click on his/her name.

[2] Account Settings: This module provide a panel to change users profile and password information. 2.2 Payment Settings: This modules helps the user to store its card information with us for easy pay and order functionality.

[3] Wallet Settings: In this module user can maintain its e-cash by adding more cash or purchasing items without using any card information. User can add cash from saved cards, or use other than save card but user can add maximum of \$1000 at one transaction and overall \$10000 are allowed. At the same time user can check for its wallet activities and details on the same page. 2.4 Order History: This feature helps the user to check all the past purchases and other details such as, date of purchase, products purchased, and payment information. 2.5 Newsletters: User can manage and subscribe or unsubscribe to our neweletters. 2.6 Logout: User can successfully logout to avoid exploitation of its account.

[4] User Cart: User can add items or removed items or edit the quatity for each item (maximum 5 quantities per product is allowed). Here user can check for total payment information and proceed to checkout.

- [5] Search Product: User can search by typing product's, id, name, brand, and category in the search bar that is present on the top of the page.
- [6] Filtered Search: User can refine its search according to price range and category. This feature will be available in search results.
- [7] Category Search: User can directly search for products according to its category by selecting categories on the left top corner of the page right next to logo.
- [8] Product Information: User can check for the product information by clicking on the heading or a given view button on each product thumbnail and discover more about the product and can add it to its cart by clicking on Add to Cart button under the display picture of an item.
- [9] Others: We have implemented fake gateway where user can select the mode of payment ie saved card, new card and wallet. User can pay for the amount by any of the listed modes and also review its payment information at the bottom and then can pay and proceed to the final step which is confirmation where a confirmation message will be displayed and generate a transaction or order it (which would also be available in user dashboard).

FUTURESCOPE

Future enhancements could include:

- Integration with third-party payment gateways.
- Implementation of a recommendation system based on user preferences.
- Scalability improvements to handle a larger user base and product
- Integration of AI-based recommendation systems.

The Online Grocery Store project has strong potential for further development and innovation. Several enhancements are planned to improve the platform's functionality, performance, and user experience.

One key area is the integration with third-party payment gateways, which would provide users with more secure and diverse payment options. Additionally, expanding support for modern payment methods, including cryptocurrency, can offer more flexibility and appeal to tech-savvy customers.

To personalize the shopping experience, an AI-based recommendation system could be implemented. This feature would analyze user preferences and shopping history to suggest relevant products, increasing user satisfaction and engagement.

CONCLUSION

The Online Grocery Store project has successfully created a user-friendly and efficient platform for online grocery shopping. It offers a smooth and convenient experience, allowing users to browse, select, and purchase groceries from the comfort of their homes.

The application uses modern web technologies to ensure fast performance, a responsive design, and secure transactions. Real-time updates on product availability and order tracking enhance the reliability and transparency of the service.

One of the key strengths of the platform is its intuitive interface, which makes navigation easy for users of all ages. The integration of secure payment systems also ensures customer data is protected throughout the transaction process.

Looking ahead, the project plans to focus on scalability and feature enhancement. Future updates may include cloud-based infrastructure, AI-based recommendations, multilingual support, subscription services, and smart device integration.

REFERENCES

Links:

- [1] <https://www.statista.com/topics/1915/online-grocery-market/>
- [2] <https://www.ibisworld.com/united-states/market-research-reports/online-grocery-sales-industry/>
- [3] <https://nielseniq.com/global/en/insights/analysis/2015/the-future-of-grocery/>
- [4] <https://www.insiderintelligence.com/content/us-online-grocery-sales-forecast-2023>

Books:

- [1] The Everything Store – Brad Stone Focuses on Amazon’s evolution, including their strategy in online groceries.
- [2] Digital Marketing Analytics – Chuck Hemann & Ken Burbary Explores how to interpret digital data for understanding consumer behaviour
- [3] Contagious: How to Build Word of Mouth in the Digital Age – Jonah Berger Useful for understanding viral trends and consumer purchasing behaviour.

Apps:

- [1] Instacart – Online grocery delivery and shopping platform.
- [2] Amazon Fresh – Grocery shopping within the Amazon app.
- [3] Walmart Grocery App – Order groceries for pickup or delivery.

CHAPTER-4

USE-CASES

Use Case 1: User Login

Use Case ID: UC001

Use Case Name: User Login

Actor(s): End User

Preconditions:

- [1] User is on the homepage
- [2] Login button is visible

Main Flow:

- [1] User clicks the "Login" button.
- [2] System redirects to the Login Page.
- [3] User enters valid credentials.
- [4] System verifies credentials.
- [5] System redirects to the User Dashboard.

Alternate Flow:

- [1] If credentials are invalid, the system displays an error message.

Postconditions:

- [2] User is authenticated and logged into the system.

Use Case 2: User Registration with Photo Upload

Use Case ID: UC002

Use Case Name: User Sign Up

Actor(s): New User

Preconditions:

- [1] User is on the homepage
- [2] "Sign Up" button is visible

Main Flow:

- [1] User clicks the "Sign Up" button.
- [2] System displays the registration form.
- [3] User enters username, password, and uploads a photo.
- [4] User clicks "Submit".
- [5] System validates inputs and photo.
- [6] Account is created and user is redirected to confirmation/dashboard.

Alternate Flow:

- [1] If validation fails, error messages are shown.

Postconditions:

- [2] New user account is created.

Use Case 3: Start New Session

Use Case ID: UC003

Use Case Name: Start New Session

Actor(s): Logged-In User

Preconditions:

- [1] User is logged in
- [2] "New Session" button is visible

Main Flow:

- [1] User clicks the "New Session" button.
- [2] System initiates a new session.
- [3] Session ID and details are displayed.
- [4] User is redirected to the session page.

Postconditions:

- [1] A new session is started successfully.

Use Case 4: Clear User History

Use Case ID: UC004

Use Case Name: Clear History

Actor(s): Logged-In User

Preconditions:

- [1] User is logged in
- [2] History exists
- [3] "Clear History" button is visible

Main Flow:

- [1] User clicks "Clear History".
- [2] System prompts for confirmation.
- [3] User confirms action.
- [4] System deletes history and displays a confirmation message.

Alternate Flow:

- [1] If user cancels, history remains unchanged.

Postconditions:

- [1] User history is cleared (if confirmed).

Use Case 5: Prompt for Legal Advice with Filtering

Use Case ID: UC005

Use Case Name: Prompt Legal Advice

Actor(s): Logged-In User

Preconditions:

- [1] User is logged in
- [2] Prompt panel is visible

Main Flow:

- [1] User opens the Prompt panel.
- [2] User inputs a legal question.
- [3] System filters the input.
- [4] If valid, legal advice is returned.
- [5] If inappropriate (foul or hate speech), system displays a warning message.

Postconditions:

- [1] User receives a valid response or a warning if the input violates content policies.
- [2] Let me know if you want this formatted as a downloadable document (PDF/Word), or if you want more use cases!

STUDENT PROFILE/RESUME

NAME:- AJIT VITTHAL ALAGUNDI

E-MAIL:- ajitalagundi@gmail.com

DIPLOMA BRANCH:- Computer Science

CONTACT NO:- 8431832849

COLLEGE:- K.L.E SOCIETY'S C. B. KORE POLYTECHNIC, CHIKODI

ADDRESS:- Chikodi - 591201

AGGREGATE:- {[CGPA:-9.22],[%Conversion:-84.70],[Results:-Distinction],[Credit Earned:-92]}

Career Objectives:-

I am seeking for an opportunity to start my career with your organization which will make me to grow competitively, which will enhance my growth by meeting social concern.

Educational Qualification:-

a) Diploma in Computer Science And Engineering:-

SEM	INSTITUTE/BOARD	YEAR OF PASSING	CLASS OBATAINED	PERCENTAGE%/SGPA
6 TH	339 – [K L E SOCIETY'S C B KORE POLYTECHNIC CHIKODI]			
5 TH	339 – [K L E SOCIETY'S C B KORE POLYTECHNIC CHIKODI]	Nov/Dec-2024	Distinction	10.00
4 TH	339 – [K L E SOCIETY'S C B KORE POLYTECHNIC CHIKODI]	Apr/May-2024	Distinction	10.00
3 RD	339 – [K L E SOCIETY'S C B KORE POLYTECHNIC CHIKODI]	Nov/Dec-2023	Distinction	9.77
2 ND	339 – [K L E SOCIETY'S C B KORE POLYTECHNIC CHIKODI]	Apr/May-2023	Distinction	9.55
1 ST	339 – [K L E SOCIETY'S C B KORE POLYTECHNIC CHIKODI]	Nov/Dec-2022	First Class	6.89

Software efficiency:-

Language	Java, Python, Css
Packages	Cad, It Skill, FEEE

Project profile:-

Title:- Purpose of Jobs

Abstract:- Jobs provide opportunities for personal growth, economic stability, and societal progress. They enable individuals to apply their skills, foster innovation, and promote financial independence

Personal profile:-

Name : Ajit Vitthal Alagundi

Father name : Vitthal

Mother name : Mallavva

Nationality : Indian

DOB : 03/06/2006

Hobbies : Reading, Writing

Language known : Kannada, English, Hindi

Marital status : Single

Alternate contact no : 8197140816

Declaration:-

I hereby informed that above mentioned details are true the best of my knowledge.

Place:- Kabbur

Ajit Vitthal Alagundi
(student name)

PHOTO GALLERY

APPENDICES

Appendix No.	Title	Description
Appendix A	Source Code	Contains the complete source code used in the project.
Appendix B	Project Screenshots	Screenshots of various modules and interfaces developed.
Appendix C	Survey/Questionnaire (if any)	Blank and filled survey forms used for data collection.
Appendix D	User Manual / Installation Guide	Step-by-step instructions for installation and usage of the application.
Appendix E	Additional Charts / Graphs	Extra data visualizations or performance results.